Chapter 7

GEOEDITOR

GeoEditor: Creating Complex Device Structure with Ease
Version 1.60

GeoEditor is friendly and powerful 2D graphical drawing software. It is designed to work with the .geo files used in Crosslight simulators LASTIP, PICS3D and APSYS and Procom. It enables the user to accurately generate and modify device structures using visual means.

7.1 Introduction

We will explain a few basic concepts for those using GeoEditor for the first time. Throughout this document, we will use [ ] to denote menu buttons and options.

- **polygon**: The basic building blocks of a complex device geometry. We are limited to only quadrilaterals (including rectangles) and triangles. All device structures must be subdivided into triangles and quadrilaterals with corner points connecting to each other. It is incorrect to connect a corner point of one polygon with an edge of its neighbour (see Figure 7.1: Requirement for Connectivity for Polygons). In general, for the purpose of better simulation convergence, we prefer quadrilaterals rather than triangles. Whenever possible, try to use rectangles and to avoid quadrilaterals with two adjacent angles greater than 90 degrees.

- **Coordinate**: Positive X direction points to the right and positive Y direction points upward.

The next section is our tutorial with an example device structure of moderate complexity. The following two sections demonstrate GeoEditor with two example structures. Walking through these examples should give you a good idea about how

GeoEditor should be used. The final section details all the menus of this software.

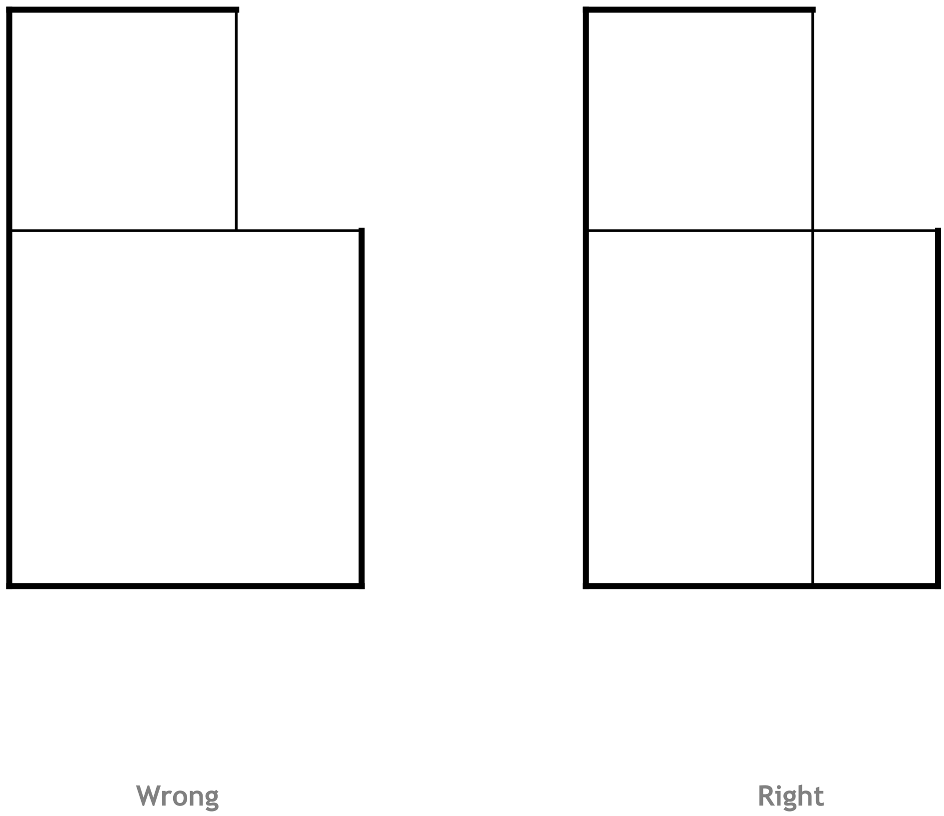

Figure 7.1: Requirement for Connectivity for Polygons

7.2 A Simple Semiconductor Structure

This is the simplest example of a structure that consists of one piece of a uniformly doped semiconductor (AlGaAs) with two contacts, one on top and the other one at the bottom (see Figure 7.2: Schematic of the simple device structure). We choose this simple example to illustrate the basic steps of using the GeoEditor so that the user does not get distracted by complicated device structures.

Please follow these steps:

1. Start the GeoEditor and left-click on the [Rectangle(R)] on the Toolbar.

2. Refer to Figure 7.2 and left-click points 1, 2 and 3, in that order. A property dialogue sheet (or property Dialogue) should pop up immediately after this step.

3. In the property Dialogue, set the desired size info [Size information]. For example, left-click the Height and set it to be 2 um. See Figure 7.3: Size Information

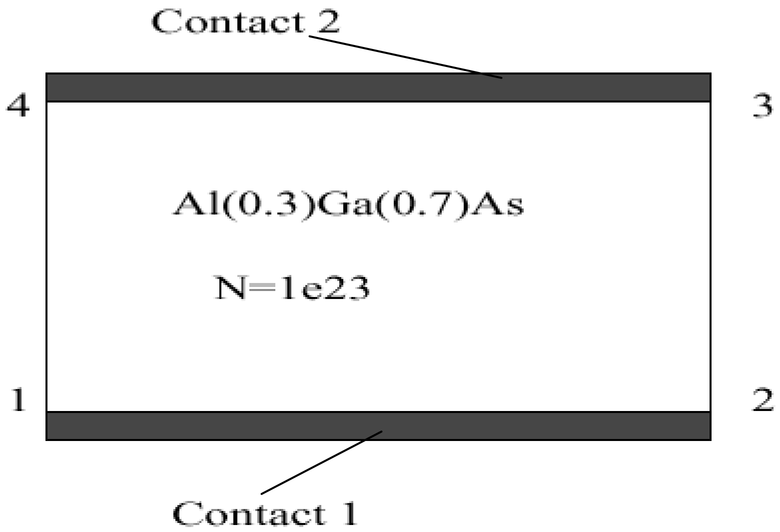

Figure 7.2: Schematic of the simple device structure

You may input the height of the layer in "Height" box. Also you can modify the width of the layer by inputting a real number in "Width" box. The names of the buttons on the bottom of the dialog depend on your operation system. In English operating system they are, "ok", "cancel", "apply", "help" accordingly. But the functions are the same despite different OS.

4. Left-click on [Mater Info] on the top of the Dialogue to define the material. The default, algaas is already displayed in the list box. Click on [Show Load macro Param] to set X=0.3 (X=0.3 in Al(X)Ga(1-X)As). Click on OK to confirm. See Figure 7.4
Choose bulk material or active material and specify the composition parameter for material macro.

5. Left-click on [Doping Info] on the top of the Dialogue to set the doping. Select N-type, and maximum concentration to be 1.e23 (m^{-3}). The edge profiles here indicate the Gaussian tail deviation outside of the edges of the rectangle. See Figure 7.5

6. Left-click on [Contact Info] on the top of the Dialogue to set the contact. Left double click the cell crossed by [Contact 1] and [Side] ,then Left-click the little black triangle and select

[1-2]. The position of the contact relative to this edge is set using From=0 To=99. Here a large number of 99 μm is used to indicate the full extension of Side 1-2 used as contact. Click on [OK] to confirm. Use similar procedure to add another contact to Side 3-4.

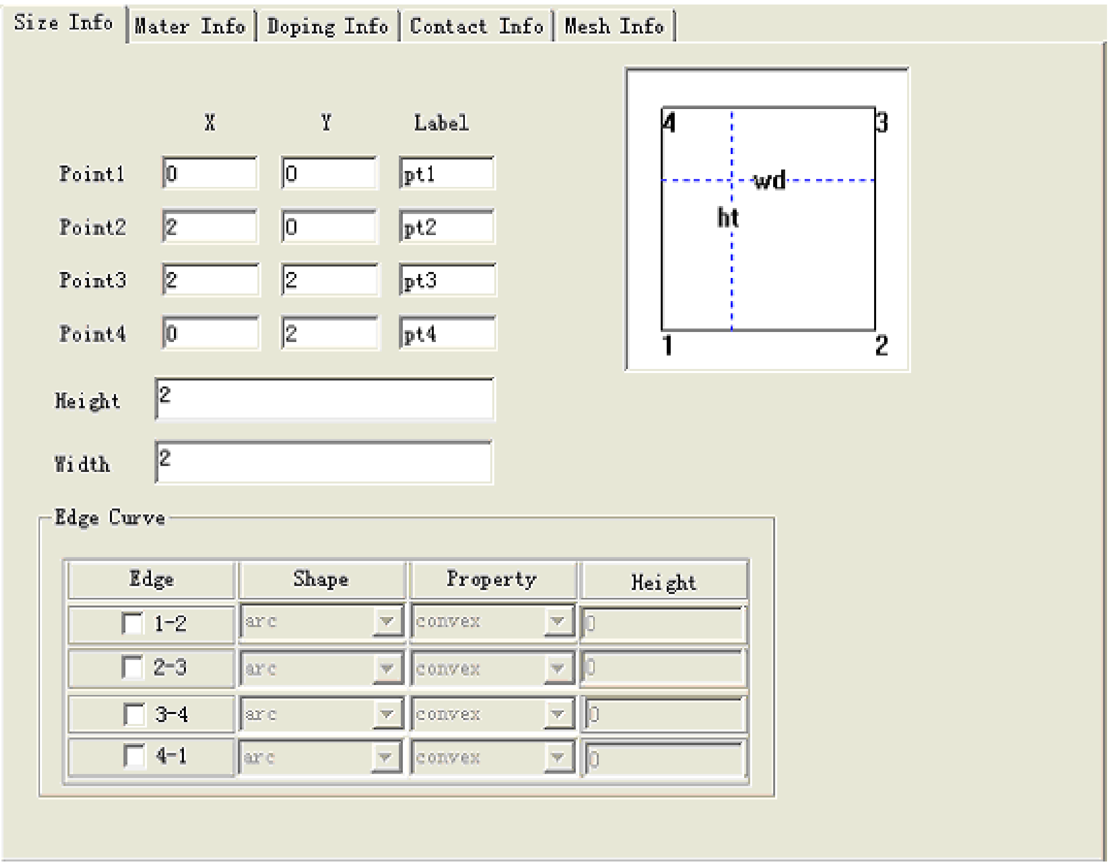

Figure 7.3: Size Information

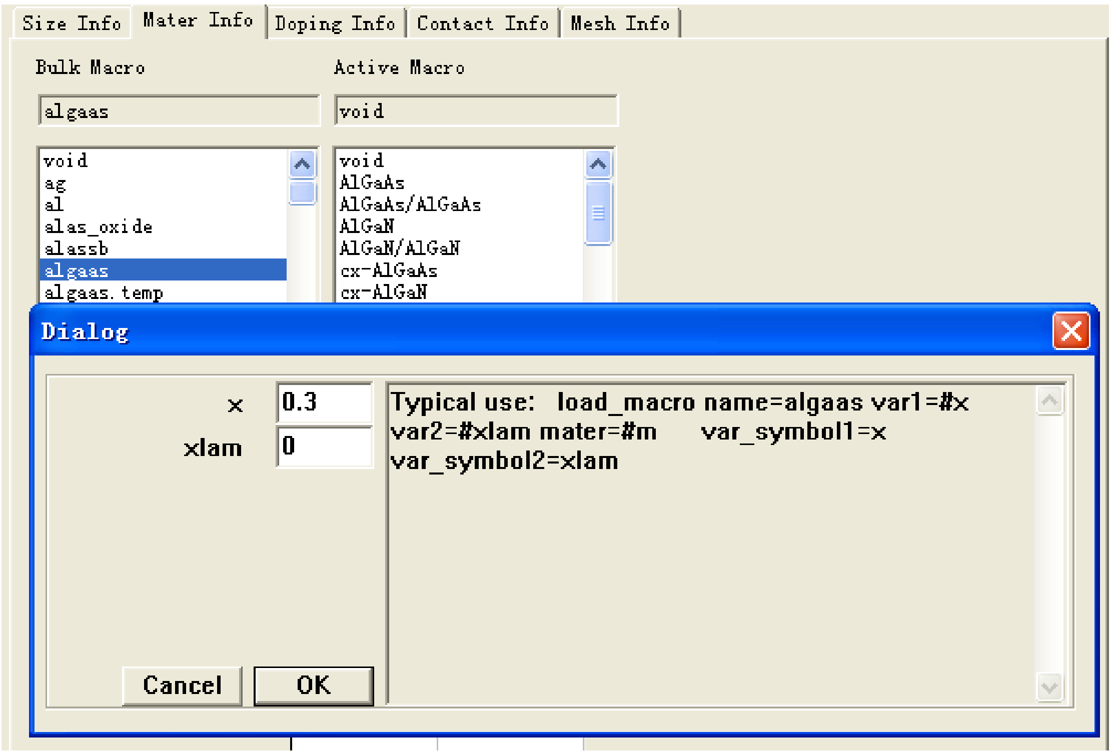

Figure 7.4: Material Information

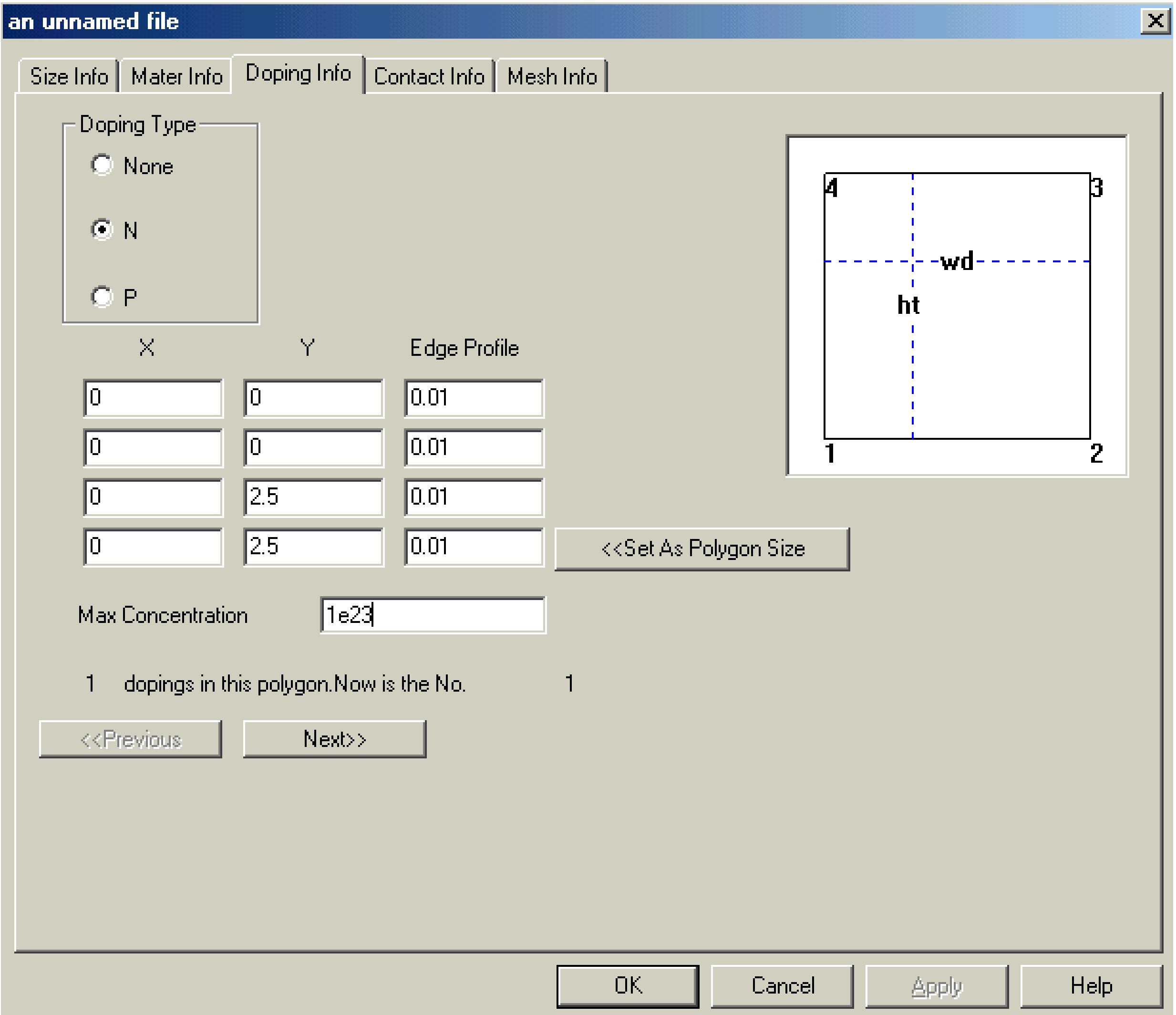

Figure 7.5: Doping Information

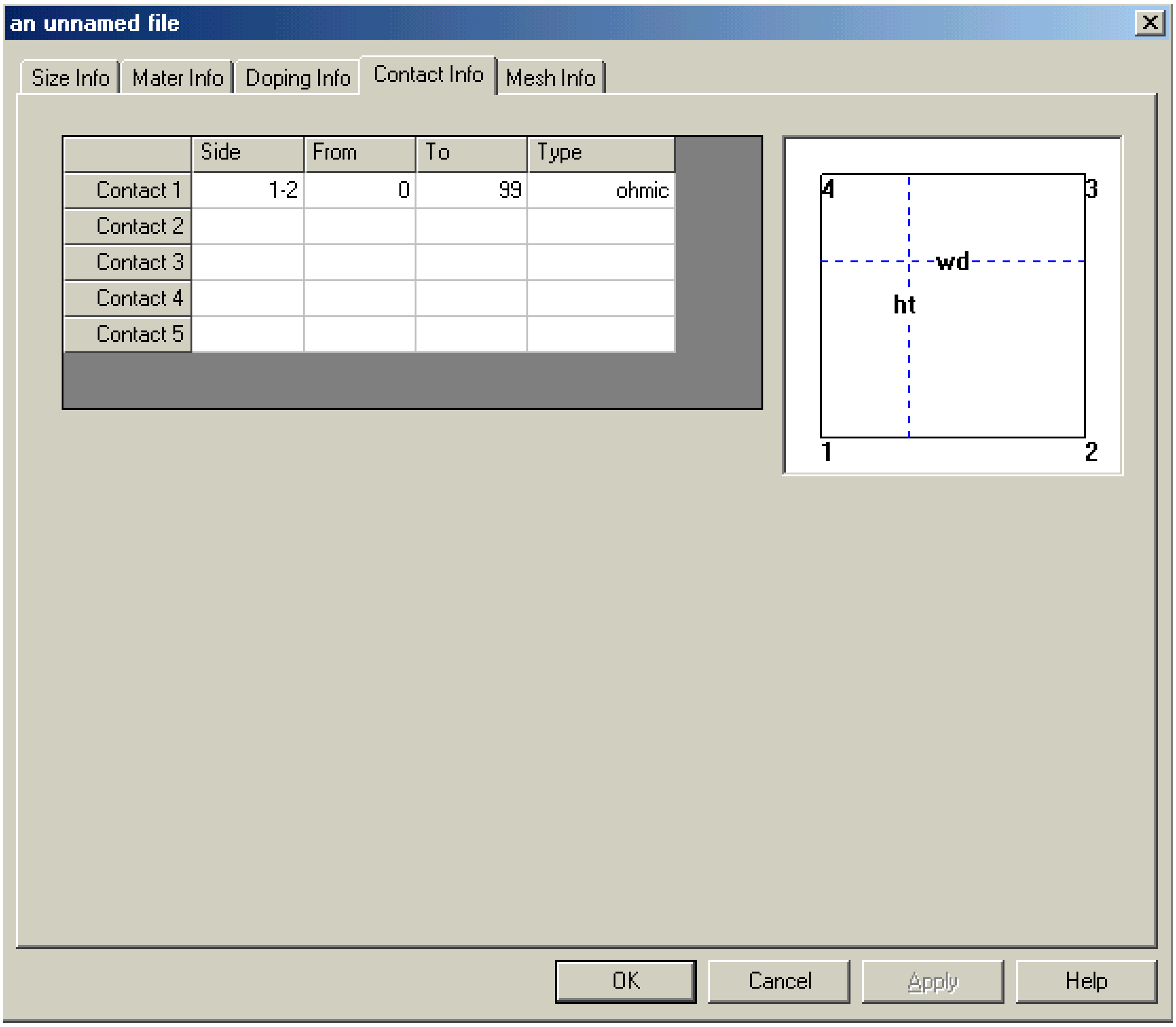

Figure 7.6: Contact Information

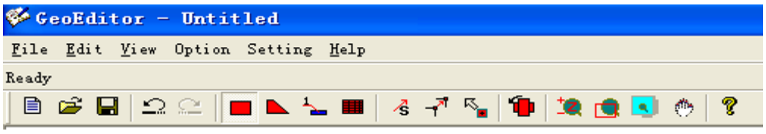

Figure 7.7: Toolbar and Menu

7. We have so far finished setting the material properties. Click on [OK] on the lower part of the property dialogue. The property dialogue will disappear and a colored rectangle will appear representing a semiconductor piece of material we just defined.

8. Left-click on the [Auto Mesh(A)] button on the Toolbar (with titled Auto Mesh symbol, See Figure 7.7) The red button with titled Auto Mesh symbol at the mid part of the toolbar is the short-cut button for [Auto Mesh]. A dialogue box will appear prompting for the total number of mesh lines in the X and Y direction you wish to use. Select 5 and 15 for the X and Y directions, respectively. Click [OK] to confirm. At field "max. spacing between X meshes" and "max. spacing between Y meshes," let them be the default value: 0.5. (Refer to Figure 7.8)

9. Finish the GeoEditor application using [File->Save As] and [File->Exit]. In the directory you have chosen, you will see four files with extension .geo, .mater, .doping and .mplt. The first, .geo file, can be used to generate the mesh for device simulation. .mater and .doping may be directly included in the .sol file to run the main simulator. The last, .mplt, is used to plot mesh of the device from .msh file, which can be generated from .geo file.

7.3 Example: A Non-regular Device

We wish to setup the simulation for a non-regular laser device structure as shown in Figure 7.9: Schematic of the Complex Device For passive devices, the procedures are similar. To simplify our description in this tutorial, we only require accurate specification of the thickness of the three material layers near the centre of the device. In real device application, all dimensions can be accurately specified.

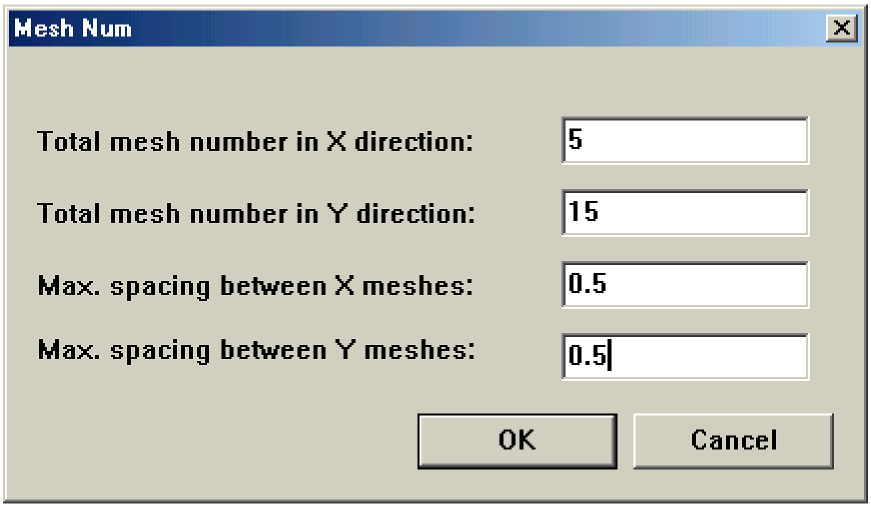

Figure 7.8: Set Mesh Information

7.3.1 Quick steps for drawing device

Please use the following procedures to quickly generate the desired geometry without setting the exact dimensions and detailed material definition. We will modify the dimensions and material definitions later.

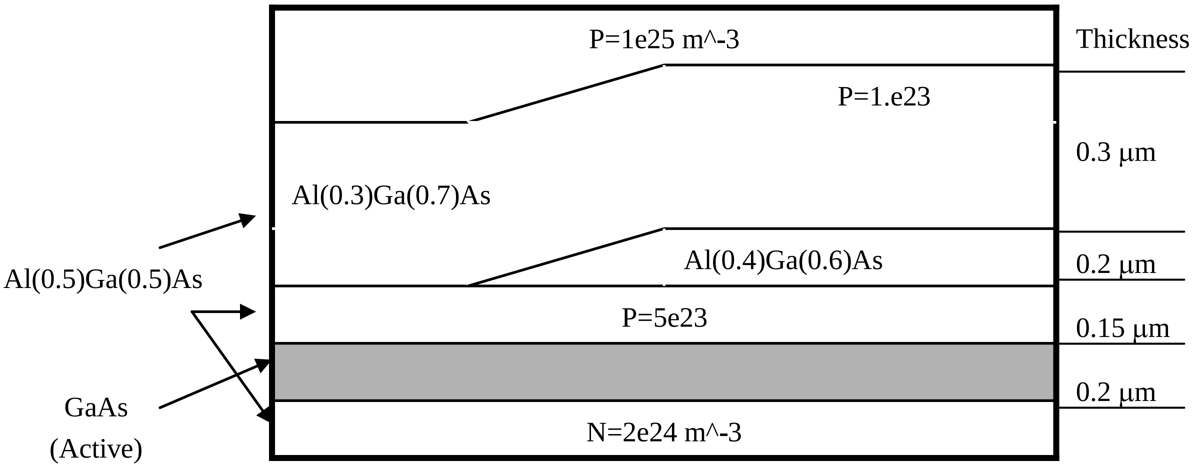

Figure 7.9: Schematic of the Complex Device

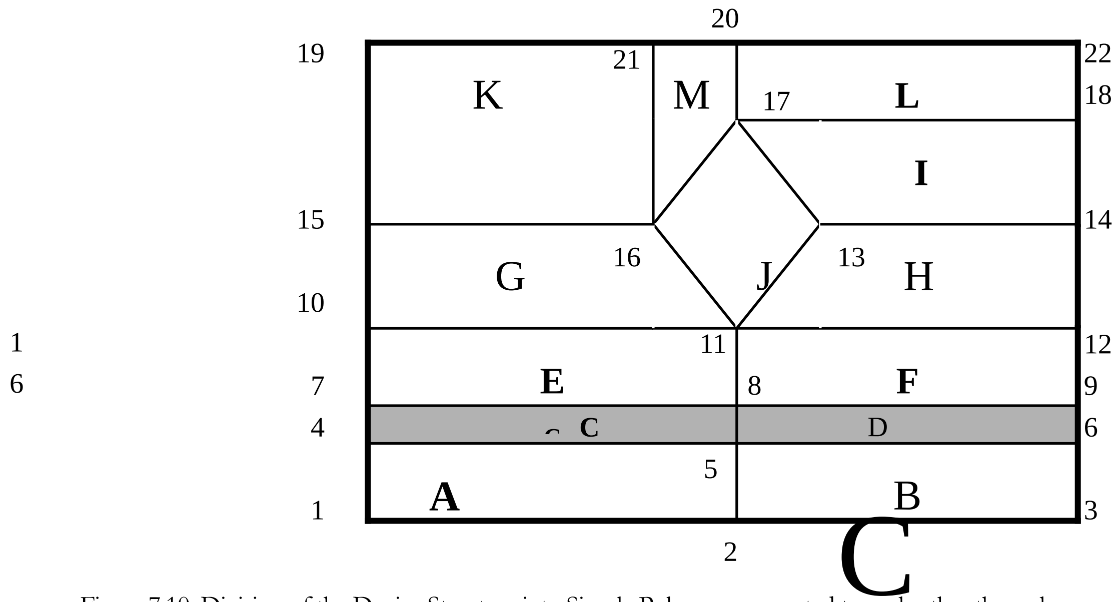

Figure 7.10: Division of the Device Structure into Simple Polygons connected to each other through corner points

1. Subdivide the device into a collective of quadrilaterals as shown in Figure 7.10: Division of the Device Structure into Simple Polygons connected to each other through corner points We will draw each polygon in the order ABCDE ... For convenient reference, we label each corner point with integers 1,2,3...

2. Run GeoEditor.

3. Click on [Setting / Grid Size] menu to set the canvas grid units. The default is 0.5 μm per grid if you skip this step.

4. Press[Rectangle(R)]. Move the mouse cursor near the lower left corner of the canvas and use the left mouse button to select point 1. Drag to the right and click for point 2. Move upwards to draw points 5 and 4 to generate polygon A. At this moment, a property sheet will pop up and give you a chance to modify the size and material properties of this newly generated polygon. However, we decide not to modify anything and click [OK] to continue.
5. Click 5, 2, and drag rightwards to produce B.

6. Click 4, 5, and drag upwards to produce C; click 5, 6, and drag upwards to produce D.

7. Use similar procedure to generate E and F.

8. Click 10, 11, and drag upwards to produce a rectangle G on top of E. Left-click the [Move vertex(V)] shortcut in the Toolbar, Click on point 16 and drag it slightly leftwards. Now the rectangle becomes a non-rectangle quadrilateral G.

9. Left-click [Rectangle(R)] again. Click 11, 12, and drag upwards for rectangle H. Left-click [Move Vertex(V)], Click on point 13 and drag rightwards to convert the rectangle into a non-rectangle quadrilateral.

10. Left-click [Rectangle(R)] again. Use similar procedure to generate quadrilateral I on top of H.

11. Click the [Quadrilateral(Q)] menu for drawing a non-rectangle quadrilateral. Click 16, 11, 13 and 17 to produce J.

12. Select Rectangle Drawing. Click 15, 16, and drag upwards for K.

13. Click 17, 18, and drag upwards for L.

14. Click the [Quadrilateral(Q)] menu for drawing a non-rectangle quadrilateral. Click 20, 17,16 and 21 to generate the last polygon of the structure, M.

So far, we have generated a device schematic with the same default material everywhere. We are to improve in the following subsections.

7.3.2 Setting the exact dimensions

We demonstrate how to set the exact dimensions in this subsection. The editing function of GeoEditor is very powerful. You can change the position of each point or set the thickness or width of each polygon. For simplicity, we are only concerned with the accurate layer thickness of the three layers near the centre of the device. Similar procedures can be used to set the widths or points.
To help us focus on geometry, we ignore the material definition here.

Start the geometry modification from polygon E as follows:

1. Click the right mouse button (right-click) to select polygon E. A dynamic menu will pop up after the right-click. Select [property] and a property sheet will pop up with size information on top. Enter the height of 0.15. Click [OK].

2. Right-click a polygon F, select [align left].

3. Use similar procedure to handle C and D to set the thickness of 0.2.

4. Right-click G / [property]. Set the thickness of 0.3 .

5. Use similar procedure to set the thickness for H and I.

7.3.3 Setting the materials

Setting the material is relatively straightforward once the geometry is defined. It is convenient to use the material copy function of the GeoEditor. This can be achieved by pressing the left mouse button and dragging from the source polygon to the target polygon. The following procedure may be used.

1. Right-click on a polygon A / [property] /[material]. Set the bulk macro to algaas with parameter 0.5 (Al composition). Click [OK] to confirm.

2. Select [Material Dragging(D)] in the Toolbar, Left-click A and hold down the mouse button. Drag rightwards to copy the material properties from A to B.

3. The same techniques can be used to set the material for C and D; E and F; G, J and I; K, M and L. Please note that C and D are active regions and we must set the active layer macro for them.

7.3.4 Setting the doping

1. Right-click a polygon A / [property] / [Doping Info]. Choose n-type doping and set the maximum concentration to be 2e24. Click [OK] to confirm.

2. Use similar procedure to set doping of polygon B, C, D, E, F, G, H, I, J, K, L, M.

7.3.5 Specifying the contacts

Our next step is to put contacts on the top and bottom of the device. The location of the contact is measured relative to the outer edge of the polygon in anti-clockwise direction. See Figure 7.11: How a Contact Segment or a Mesh Point Array is specified on edges of a Polygon for further explanation. Use the following steps for the contacts.

Here two numbers r1 and r2 are measured from point 1 for the lower edge to define a contact. For the upper edge, q1 and q2 are measured from point 3. Similarly for other edges, the direction goes anti-clockwise.

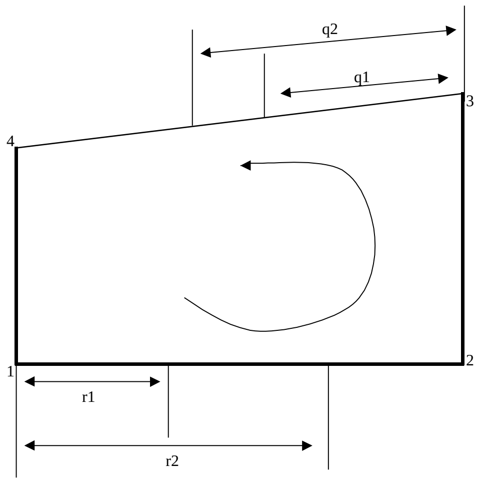

Figure 7.11: How a Contact Segment or a Mesh Point Array is specified on edges of a Polygon

1. Right-click a polygon A / [property] / [Contact Info]. Choose the lower edge(1-2) and set the location of the contact relative to this edge(e.g. From=0.0, To=99.0). Please note that you can specify a large number for the [to] value if the contact extends to the end of the edge.

2. Use similar procedures to set the contact at the bottom of
B, on the top of K and M. Note that only a part of the upper edge of M is covered by the contact.

7.3.6 Mesh generation and save drawing

As the final step, we will generate the mesh. There are two ways to do this. The first is to let GeoEditor generate the mesh automatically, based on distribution of doping and material discontinuity. The second is to set the mesh manually. For the 2nd approach, we only need to put mesh on the outer edges of the polygons located in the peripheral areas. The mesh lines will automatically propagate to neighbouring polygons.

Choose the first approach to let GeoEditor automatically generate the mesh. To do so, select [Auto Mesh(A)]. At this moment, GeoEditor will prompt for the total number of mesh points in x and y directions. Enter typical values 15 and 30 for x and y directions, respectively. Based on the total mesh, GeoEditor will decide where to allocate the mesh lines.

We need to save the drawing related information in files. Using [save as] we choose a base file name, say test1, and GeoEditor will save the information in test1.mplt, test1.geo, test1.mater and test1.doping; test1.geo, test1.mater, test1.mplt and test1.doping are ASCII input files acceptable by our simulators directly.

7.4 GeoEditor Reference

7.4.1 Menu

The menu functions are detailed in Table 7.1: Menu Part I and Table 7.2: Menu Part II. Some important menus are represented by menu bar items and listed on the first column in the tables.

Tool bar	Menu	Function
	File/New	Create a .geo file
	File/Open	Open a .geo file
	File/Close	Close a .geo file
	File/Save	Save the current .geo file
	File/Save as	Save the current .geo file as a new name.
	File/Exit	Exit program
	Edit/Undo	Undo the change
	Edit/Redo	Redo the change
	Edit/Select All	Select all layers
	Edit/Copy	Copy layer(s) or column(s)
	Edit/Paste	Paste layer(s) or column(s)
	Edit/delete	Delete selected polygon.
	Edit/Polygon property	Show selected polygon property
	View/toolbar	Show or hide toolbar
	View/status bar	Show or hide status at the top board
	View/Doping	Show doping or geometry of the device or Geometry.

	View/Centre	Change view point to centre

Table 7.1: Menu Part I

Tool bar	Menu	Function
[Create Remark]	Option/Remarks	Create, modify or delete remarks
[Rectangle(R)]	Option/Rectangle	Draw rectangle
[Quadrilateral(P)]	Option/Quadrilateral	Draw quadrilateral
[Auto Mesh(A)]	Option/Auto mesh	Apply the "auto mesh" function
[Select(S)]	Option/Select	Select Polygon(You can use Ctrl & Shift)
[Move Vertex(V)]	Option/Move Vertex	Move the vertex
[Material Dragging(D)]	Option/Material Dragging	Copy material by dragging.
[Rotate(O)]	Option/Rotation	Rotate the device 90 degree anti-clockwise.
[Zoom(Z)]	Option/Zoom	Zoom the device
[Zoom Center(F)]	Option/Zoom Center	Display the device in actual size and zoom center.
[Zoom Window(W)]	Option/Zoom Window	select an area for zooming
[Move All(M)]	Option/Move all	Move the whole device
		the selected polygon.
	Option/Horizontal Mirror	Generate a horizontal mirror of the device.
	Option/Vertical Mirror	Generate a vertical mirror of the Device.
	Option/ Change Mater Num	Change material number of
	Option/Set x_range and y_range	Set x_range and y_range parameter.
	Setting/Zoom Drection	zoom in (out) the device in X, Y, or both directions
	Setting/Back Clour	Set the background colour
	Setting/Contact Clour	Set the colour of the contact
	Setting/Grid size	Set the grid size

Help/About GeoEditor	Version info of GeoEditor
Help/Geoeditor Help	GeoEditor help files.

Table 7.2: Menu Part II

7.4.2 Basic operation of GeoEditor

1. Drawing polygons.

a) draw rectangle. Left-click to select the first and second points. The third point determines the height of the rectangle.

b) draw quadrilateral (triangle). Left-click to select all the points. You can press the SHIFT when you are drawing the quadrilateral. In this mode, you can draw vertical or horizontal edges . A property sheet will pop up after the drawing to allow any modifications of the newly created polygon.

c) the following situations are not allowed:

i) The first edge of a new polygon does not coincide with an edge of any existing polygon.
ii) A new polygon overlaps with any existing polygons.
iii) The new polygon has an internal angle greater than 180 degrees.

2. Use [Setting / Grid Size] to set the ruler grid size in the drawing canvas.

3. Edit the properties of a polygon by right clicking / [property]. Or double-right-click on the target polygon.

4. Zoom: Zoom in/out the display, not the actual device size. There are two ways:

i) Select [Option/Zoom] to zoom the device.
ii) Use [Option/Zoom Window] to select an area for zooming.

5. Use left mouse key drag and drop to copy the material and doping properties between two existing polygons.

6. Use [Edit/Copy] to copy a polygon. press [Edit/Copy], use left key to drag the polygon you wish to copy and drop it in a new location. Please note that the new polygon must have one edge coincide with an existing polygon. Otherwise, this operation will be ignored.

7. Select [Move Vertex(V)] tool. Then, move a corner point of a polygon. Use left mouse button to drag any one of the point squares to a new position. This operation will affect its neighbors and its doping.

8. Align polygons. Use [Align]/[Align Left] to align the polygon to its neighbour on the left. The upper and lower edges will become horizontal. Similarly for [Align]/[Align Right], [Align]/[Align Top] and [Align]/[Align Bottom].

9. To centre the drawing in the canvas, press [View]/[Centre].

10.To center the drawing and to show the whole device in the screen, press[option]/[Zoom Center].

11. Mesh generation. There are two ways of generating the mesh.

i) In property sheet/mesh, set the mesh in the outer edge. You need to specify which edge, the position (from/to), the mesh number and mesh ratio. The related parameters correspond to put_mesh statement and are explained in more detail in the Reference Manuals of the respective products.

ii) Automatic mesh generation. If you select [Auto mesh], you will be prompted to enter the total mesh number in x and y direction. GeoEditor will determine how the mesh lines should be distributed.

7.4.3 Polygon properties

When you right click on a polygon and clicking [property], a property sheet dialogue box will pop up displaying all parameters related to the polygon. A polygon is completely defined by the following parameters:

1. size information: You can either define the absolute coordinates of each point, or specify the height or width of the polygon. And you can also define the edge curve.

2. material: You can set the bulk macro, active layer macro (for active layer or quantum wells only), and any composition gradings. More details about material macros can be found in the User's Manual.

3. doping: You can set the type of doping, the maximum uniform doping within the polygon and the Gaussian tail standard deviation (or edge_prof for edge profile). More explanation can be found in the product's Reference manual under statement doping.

4. contact: Used to specify the type, the edge and position of a contact. Details are explained in the Reference Manual regarding statement polygon and contact.

5. mesh: There are two ways of generating the mesh.

i) set the mesh in the outer edge. You need to specify which edge, the position (from/to), the mesh number and mesh ratio. The related parameters correspond to

put_mesh statement and are explained in more detail in the Reference Manuals.

ii). Automatic mesh generation: you will be prompted to enter the total mesh number in x and y direction. GeoEditor will determine how the mesh lines should be distributed.

7.5 Typical Examples

Several typical examples provided with the core products of LASTIP are given to demonstrate how to use GeoEditor to prepare input files. The demonstrations will show the procedures step by step so that the user can follow with ease.

7.5.1 Example 1: A simple quantum well laser for LASTIP

This example is provided in the path: ~\ lastip_examples \A_layered_LD

This is a single-column structure with material grading.

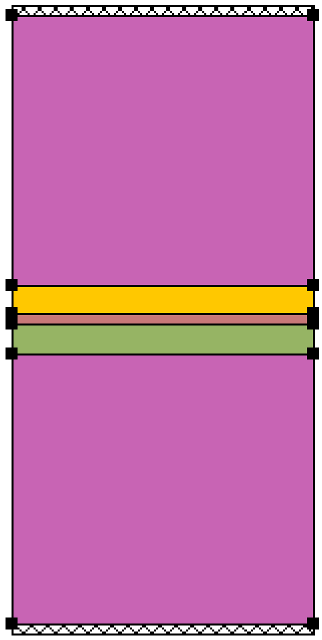

Figure 7.12: Example gaas 10

1. Create the first polygon of the device

    1. Option --> Rectangle Drawing: Left-click three points to define a polygon

    2. Input size and material information in popup dialogue: Size Info --> Height: 1.35, Width: 1.5
    Mater Info --> Bulk Macro: algaas
    Show Load Macro Param:x= 0.71

2. Create the second layer with material grading

    1. Option --> Rectangle Drawing: Left-click the two top points of the first polygon and left-click a point upward

    2. Input size and material information in popup dialogue. Size Info --> Height: 0.1462
    Mater Info --> Bulk Macro: algaas
    Material Grading: x
    Direction: y
    Distance: 0 0.1462
    Composition: 0.71 0.33

Attention: The coordinates used here are relative to the left and bottom points of this polygon.

3. Create the third polygon---active region of the device

    1. Option --> Rectangle Drawing: Left-click the two top points of the second polygon and left-click a point upward

    2. Input size and material information in popup dialogue: Size Info --> Height: 0.0076
    Mater Info --> Bulk Macro: gaas
        Active Macro: AlGaAs/AlGaAs
        Show Active Macro Param --> xw: 0.
    xb: 0.33

4. Create the fourth layer through copy and paste function

    1. Option --> Select: click the second polygon
    Edit --> Copy
    Edit --> Paste: click the left top point to paste a polygon on the third one directly

    2. Change the material information: Double click this polygon and in popup dialogue:
    Mater Info --> Bulk Macro: algaas
    Material Grading: x
    Direction: y
    Distance: 0 0.1462
    Composition: 0.33 0.71

5. Create fifth polygon by copying the bottom polygon

6. Add doping information
    Choose the bottom polygon and double-click it:
    Doping Info --> Doping Type: N
        --> Set As Polygon Size
            --> Max. Concentration: 1.e24
    Choose the top polygon and double-click it:
    Doping Info --> Doping Type: P
        --> Set As Polygon Size
            --> Max. Concentration: 1.e24

7. Add contact information: Choose the bottom polygon and double-click it:
    Contact Info --> Contact 1 --> side: double click the blank and choose 1-2
        --> from: 0
        --> to: 1.5
    Remember that if the length of the contact is longer than that of the side, the redundant

length is properly ignored.

Type: double click the bland and choose ohmic.

The top contact can be created in the same way along edge 3 – 4 of 5th polygon.

8. Add mesh information. Choose the bottom polygon and double-click it:

Mesh Info --> 1st line --> Side: double click the blank and choose 1-2
--> From: 0
--> To: 1.5
--> Ratio: 1
--> Mesh No: 2

Repeat the steps to add all mesh information.

9. Add some information that cannot be edited by GeoEditor. Users must revise the .geo, .mater and .doping files manually.

The formed gaas10.geo is listed in the following:

```
begin_geometry
point   label=a001 xy=[   0.000000E+000   0.000000E+000 ]
point   label=a002 xy=[   0.000000E+000   0.135000E+001 ]
point   label=a003 xy=[   0.000000E+000   0.149620E+001 ]
point   label=a004 xy=[   0.000000E+000   0.150380E+001 ]
point   label=a005 xy=[   0.000000E+000   0.165000E+001 ]
point   label=a006 xy=[   0.000000E+000   0.300000E+001 ]
$
point   label=b001 xy=[   0.150000E+001   0.000000E+000 ]
point   label=b002 xy=[   0.150000E+001   0.135000E+001 ]
point   label=b003 xy=[   0.150000E+001   0.149620E+001 ]
point   label=b004 xy=[   0.150000E+001   0.150380E+001 ]
point   label=b005 xy=[   0.150000E+001   0.165000E+001 ]
point   label=b006 xy=[   0.150000E+001   0.300000E+001 ]
$
polygon name=p001 4_points=[ a001 b001 b002 a002 ] material= 1 &&
   boundary_1=[ a001 b001 ] limits_1=[     0.000000E+000   0.150000E+001 ]
polygon name=p002 4_points=[ a002 b002 b003 a003 ] material= 2
polygon name=p003 4_points=[ a003 b003 b004 a004 ] material= 3
polygon name=p004 4_points=[ a004 b004 b005 a005 ] material= 4
polygon name=p005 4_points=[ a005 b005 b006 a006 ] material= 1 &&
   boundary_2=[ b006 a006 ] limits_2=[     0.000000E+000   0.150000E+001 ]
$
end_geometry
begin_meshgen
internal_xpoint xp_size=   0.1000E-03
put_mesh polygon=p001 edge=[ b001 b002 ]   &&
```

```
   point_from=   0.000000E+000 point_to=   0.135000E+001 &&
   number=   9 ratio=   0.800000E+000
put_mesh polygon=p002 edge=[ b002 b003 ]   &&
   point_from=   0.000000E+000 point_to=   0.146200E+000 &&
   number=   6 ratio=   0.800000E+000
put_mesh polygon=p003 edge=[ b003 b004 ]   &&
   point_from=   0.000000E+000 point_to=   0.760000E-002 &&
   number=   4 ratio=   0.100000E+001
put_mesh polygon=p004 edge=[ b004 b005 ]   &&
   point_from=   0.000000E+000 point_to=   0.146200E+000 &&
   number=   6 ratio=   0.120000E+001
put_mesh polygon=p005 edge=[ b005 b006 ]   &&
   point_from=   0.000000E+000 point_to=   0.135000E+001 &&
   number=   9 ratio=   0.120000E+001
$
put_mesh polygon=p001 edge=[ a001 b001 ]   &&
   point_from=   0.000000E+000 point_to=   0.150000E+001 &&
   number=   2 ratio=   0.100000E+001
$
mesh_output mesh_outfile=gaas10.msh               order=yes
end_meshgen
```

The formed gaas10.mater is listed in the following:

```
contact num= 1 type=ohmic
contact num= 2 type=ohmic
load_macro name=algaas mater= 1 &&
   var_symbol1=x                            var1=   0.7100E+00
$
mater_var name=c2     variation=linear_y data_num= 4 &&
   4_values=[ &&
      0.13500E+01   0.71000E+00 &&
      0.14962E+01   0.33000E+00 ]
load_macro name=algaas mater= 2 &&
   var_symbol1=x                            var_name1=c2
load_macro name=gaas mater= 3
get_active_layer name=AlGaAs/AlGaAs mater= 3 &&
   var_symbol1=xw                           var1=   0.0000E+00 &&
   var_symbol2=xb                           var2=   0.3300E+00
active_reg mater= 3 &&
   thickness=   0.76000E-02
$
$
mater_var name=c4     variation=linear_y data_num= 4 &&
```

```
   4_values=[ &&
     0.15038E+01   0.33000E+00 &&
     0.16500E+01   0.71000E+00 ]
load_macro name=algaas mater= 4 &&
   var_symbol1=x                          var_name1=c4
```

The formed gaas10.doping is listed in the following:

```
$x_range=   0.000000000000E+000   0.150000000000E+001
$y_range=   0.000000000000E+000   0.300000000000E+001
$ Wave boundary is used only when optical modes are solved
wave_boundary point_ll=[   0.000000000000E+000   0.000000000000E+000] &&
   point_ur=[   0.150000000000E+001   0.300000000000E+001]
doping charge_type=donor &&
   max_conc=   0.100000E+025 shape=polygon &&
   edge1_prof=[   0.000000E+000   0.000000E+000   0.100000E-002] &&
   edge2_prof=[   0.150000E+001   0.000000E+000   0.100000E-002] &&
   edge3_prof=[   0.150000E+001   0.135000E+001   0.100000E-002] &&
   edge4_prof=[   0.000000E+000   0.135000E+001   0.100000E-002]
doping charge_type=acceptor &&
   max_conc=   0.100000E+025 shape=polygon &&
   edge1_prof=[   0.000000E+000   0.165000E+001   0.100000E-002] &&
   edge2_prof=[   0.150000E+001   0.165000E+001   0.100000E-002] &&
   edge3_prof=[   0.150000E+001   0.300000E+001   0.100000E-002] &&
   edge4_prof=[   0.000000E+000   0.300000E+001   0.100000E-002]
```

The formed gaas10.mplt is listed in the following:

```
$file:gaas10.mplt
begin_plotmesh
plot_mesh mesh_infile=gaas10.msh              &&
   plot_device=postscript
end_plotmesh
```
The construction of the device is finished.

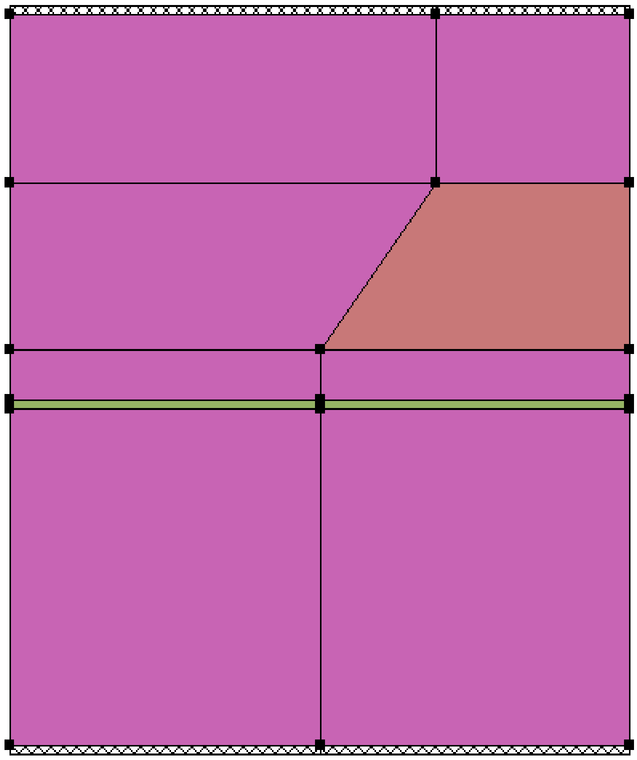

Figure 7.13: Example msas

7.5.2 Example 2: Multimode simulation (bulk self-aligned) for LASTIP

This example is provided in the path: ~\ lastip_examples \multimode\msas

The structure of this device is of two columns with two irregular polygons. First, set up the layers of the first column one by one and then set the second column by functions of 'copy' and 'paste' so that a device with only rectangles is constructed. Next, modify the corresponding polygons to fit its actual structure.

Since the functions of 'copy' and 'paste' apply only to the size and material information, define the size and material properties first. After all these are done, the doping, mesh and contact information is then added to the device.

1. Create the first polygon of the device

1. Option --> Rectangle Dawing: Left-click three points to define a polygon
2. Input size and material information in popup dialogue: Size Info --> Height: 2, Width: 2

Mater Info --> Bulk Macro: algaas, Show Load Macro Param:x= 0.45

2. Create the second polygon---active region of the device

1. Option --> Rectangle Dawing: Left-click the two top points of the first polygon and left-click a point upward
2. Input size and material information in pop up dialogue: Size Info --> Height: 0.04
   Mater Info --> Bulk Macro: algaas
   Show Load Macro Param:x= 0.15
   Active Macro: AlGaAs
   Show Active Macro Param --> xw: 0.15;

3. Setup the third polygon through copy and paste functions

1. Option --> Select: click the first polygon
   Edit --> Copy
   Edit --> Paste: click the left top point to paste a polygon on the second one directly
2. Change the size information: Double click this polygon and in popup dialogue:
   Size Info --> Height: 0.3

4. Setup the following two polygons: Repeat step 3 to setup the next two polygons with height 1.

5. Setup the second column by 'copy' and 'paste all'

Edit --> Select All
Edit --> Copy
Edit -> Paste: click the right bottom point to paste the whole column

6. Change material property of the second column. Only the fourth layer's material of the second column is different from the first column in this device. Choose this polygon and double click it. In popup dialogue:

Mater Info --> Bulk Macro: gaas

7. Modify the fourth layer of the first column from rectangle to quadrilateral. Choose this polygon and double click it. In popup dialogue:

Size Info --> point3: x=2.75

8. Modify the fifth polygon's size. Choose the fifth polygon of the first column and right-click. In popup prompt, choose

Align --> align bottom

Thus, the structure of the device is constructed with precise coordinates and material property. In the following steps, doping, contact and mesh information will be added.

9. Add doping, contact and mesh information to the left and bottom polygon. Choose the polygon and double-click it:

Doping Info --> Doping Type: N
--> Set As Polygon Size
--> 1st point: (0 0 0.01)
2nd point: (10 0 0.02)
3rd point: (10 2 0.01)
4th point: (0 2 0.02)

Remember that the coordinate here is relative to the left and bottom of this polygon.

--> Max. Concentration: 1.e23

Contact Info --> Contact 1 --> side: double click the blank and choose 1-2
--> from: 0
--> to: 2

Remember that if the length of contact is longer than that of the side, the redundant length is properly ignored.

--> Type: double click the blank and choose ohmic

Mesh Info --> 1st line --> Side: double click the blank and choose 1-2
--> From: 0
--> To: 2
--> Ratio: 0.7
--> Mesh No: 6

Thus, almost all the information of this polygon has been included.

10. Use the same way to define the doping, contact and mesh information of the left polygons. At this step, almost all the information of this polygon has been included.

11. Save file as "GeoEditor/examples/multimode/msas", and four files of msas.geo, msas.mater, msas.doping and msas.mplt are formed.

12. Modifies files for information not available by GeoEditor. There are some information that cannot be edited by GeoEditor. Users must revise the .geo, .mater and . doping files manually to add or modify some information not available in GeoEditor. For example, in this device, the user must add parameters of active_reg including exch_coef, tau_scat, mode, etc and parameters of put_mesh like extra_point manually in the corresponding msas.mater and msas.geo file .

The formed msas.geo is listed in the following:

```
$file:msas.geo
$***********GEOMETRY DESCRIPTION******************
begin_geometry
point label=a1 xy=( 0.   0.)
point label=a2 xy=( 0.   2.0)
point label=a3 xy=( 0.   2.04)
point label=a4 xy=( 0.   2.34)
```

```
point label=a5 xy=( 0.   3.34)
point label=a6 xy=( 0.   4.34)
$
point label=b1 xy=( 2.   0.)
point label=b2 xy=( 2.   2.0)
point label=b3 xy=( 2.   2.04)
point label=b4 xy=( 2.   2.34)
point label=b5 xy=( 2.75   3.34)
point label=b6 xy=( 2.75   4.34)
$
point label=c1 xy=( 4.   0.)
point label=c2 xy=( 4.   2.0)
point label=c3 xy=( 4.   2.04)
point label=c4 xy=( 4.   2.34)
point label=c5 xy=( 4.   3.34)
point label=c6 xy=( 4.   4.34)
$
$
polygon name=p1 4_points=(a1 b1 b2 a2 ) material=1 &&
   boundary_1=(a1 b1) limits_1=(0. 2.001)
polygon name=p2 4_points=(a2 b2 b3 a3 ) material=2
polygon name=p3 4_points=(a3 b3 b4 a4 ) material=1
polygon name=p4 4_points=(a4 b4 b5 a5 ) material=1
polygon name=p5 4_points=(a5 b5 b6 a6 ) material=1 &&
   boundary_2=(a6 b6) limits_2=(0.0 2.751)
$
polygon name=q1 4_points=(b1 c1 c2 b2 ) material=1 &&
   boundary_1=(b1 c1) limits_1=(0. 2.001)
polygon name=q2 4_points=(b2 c2 c3 b3 ) material=2
polygon name=q3 4_points=(b3 c3 c4 b4 ) material=1
polygon name=q4 4_points=(b4 c4 c5 b5 ) material=3
polygon name=q5 4_points=(b5 c5 c6 b6 ) material=1 &&
   boundary_2=(b6 c6) limits_2=(0.0 1.251)
$
$
end_geometry
$***********MESH GEN******************************
begin_meshgen
put_mesh polygon=q1 edge=(c1 c2) point_from=0. point_to=2.0 &&
   number=8 ratio=0.7 extra_point_2=0.0010
put_mesh polygon=q2 edge=(c2 c3) point_from=0. point_to=0.04 &&
   number=4 ratio=1.2 extra_point_1=0.0010 extra_point_2=0.0010
put_mesh polygon=q3 edge=(c3 c4) point_from=0.0 point_to=0.30 &&
   number=6 ratio=1.3 extra_point_1=0.0010 extra_point_2=0.0010
```

```
put_mesh polygon=q4 edge=(c4 c5) point_from=0.0 point_to=1.0 &&
   number=7 ratio=-1.3 extra_point_1=0.0010 extra_point_2=0.0010
put_mesh polygon=q5 edge=(c5 c6) point_from=0.0 point_to=1.0 &&
   number=5 ratio=1.3 extra_point_1=0.0010
put_mesh polygon=p1 edge=(a1 b1) point_from=0.0 point_to=2.0 &&
   number=6 ratio=0.70
put_mesh polygon=q1 edge=(b1 c1) point_from=0.0 point_to=2.0 &&
   number=6 ratio=1.30
mesh_output   mesh_outfile=msas.msh order=yes
$
end_meshgen
```

The formed msas.mater is listed in the following:

```
contact type=ohmic num=1
contact type=ohmic num=2
$ ****************************
load_macro name=algaas var1=0.45 mater=1 var_symbol1=x
load_macro name=algaas var1=0.15 mater=2 var_symbol1=x
load_macro name=gaas mater=3
get_active_layer name=AlGaAs var1=0.15 mater=2 var_symbol1=xw
active_reg type=macro mater=2 &&
    exch_coef=3.d-10 tau_scat=1.d-13    &&
    mode=te dip_factor=0.7
```

The formed msas.doping is listed in the following:

```
doping impurity=shal_dopant charge_type=donor max_conc=1.e23   &&
   x_prof=(0.0D0, 10.0D0,   0.02D0, 0.02D0 )   &&
   y_prof=(0.D0,   2.D0,      0.01D0, 0.01D0 )
doping impurity=shal_dopant charge_type=acceptor max_conc=1.e23 &&
   x_prof= 0.D0, 50.D0,   0.15D0, 0.15D0   &&
   y_prof=( 2.00,   2.04 , 0.01,   0.01   )
doping impurity=shal_dopant charge_type=acceptor max_conc=1.e24 &&
   x_prof= 0.D0, 50.D0,   0.15D0, 0.15D0   &&
   y_prof=( 2.04,   2.34 , 0.01,   0.01   )
doping impurity=shal_dopant charge_type=acceptor max_conc=2.e24 &&
   x_prof= 0.D0, 50.D0,   0.15D0, 0.15D0   &&
   y_prof=( 3.34,   4.34 , 0.01,   0.01   )
doping impurity=shal_dopant charge_type=acceptor max_conc=2.e24   &&
shape=polygon &&
    edge1_prof= (0.      2.34 0.01)    &&
    edge2_prof= (2.      2.34 0.01)    &&
```

```
    edge3_prof= (2.75   3.34 0.01)     &&
    edge4_prof= (0.     3.34 0.01)
doping impurity=shal_dopant charge_type=donor max_conc=3.e14   &&
shape=polygon &&
    edge1_prof= (2.     2.34 0.01)     &&
    edge2_prof= (4.     2.34 0.01)     &&
    edge3_prof= (4.     3.34 0.01)     &&
    edge4_prof= (2.75   3.34 0.01)
```

The formed msas.mplt is listed in the following:

```
$file:msas.mplt
$*************PLOTMESH**************************
begin_plotmesh
plot_mesh mesh_infile=msas2.msh plot_device=postscript
end_plotmesh
```

The whole device is completed.

7.5.3 Example 3: 2D ridge waveguide laser for LASTIP

This example is provided in the path: ~\ lastip_examples \mqw_ridge\GaAs\

This is a device of two columns with material grading. Generally, we build the first column one by one with size and material information and form the second column through copy and paste functions. Then, we change the properties of the second column, such as adding doping, contact and material information of the structure.

Figure 7.14: Example gaas2

1. Create the first polygon of the device

    1. Option --> Rectangle Drawing: Left-click three points to define a polygon
    2. Input size and material information in popup dialogue:
        Size Info --> Height: 1.4962, Width: 1.5
     Mater Info --> Bulk Macro: algaas
     Material Grading: x
     Direction: y
     Distance: 0 1.35 1.4962
     Composition: 0.71 0.71 0.33

2. Create the second polygon---active region of the device

    1. Option --> Rectangle Drawing: Left-click the two top points of the first polygon and left-click a point upward
    2. Input size and material information in popup dialogue:
     Size Info --> Height: 0.0076
     Mater Info --> Bulk Macro: gaas
        Active Macro: AlGaAs/AlGaAs
        Show Active Macro Param --> xw: 0
            xb: 0.33

Note that the coordinates used here are relative to the left and bottom point of this polygon.

3. Setup the third polygon through copy and paste functions

    1. Option --> Select: click the first polygon
        Edit --> Copy
        Edit --> Paste: click the left top point to paste a polygon on the second one directly
    2. Change the size and material information:
        Double click this polygon and in popup dialogue:
     Size Info --> Height: 0.2
     Mater Info --> Bulk Macro: algaas
     Material Grading: x
     Direction: y
     Distance: 0 0.1462 0.2
     Composition: 0.33 0.71 0.71
    To view the whole structure:
     Option --> Zoom Center

4. Setup the fourth polygons: Repeat step 3 to setup the next polygon

Size Info --> Height: 1.2962

Mater Info --> Bulk Macro: algaas

Material Grading: None

Show Load Macro Param:x= 0.71

5. Setup the second column by 'copy' and 'paste all'

Edit --> Select All

Edit --> Copy

Edit --> Paste: click the right bottom point to paste the whole column

Double click any polygon of the second column and in popup dialogue:

Size Info --> width: 6.5

6. Change material property of the second column. Only the fourth layer's material of the second column is different from the first column in this device. Choose this polygon and double click it.

In popup dialogue:

Mater Info --> Bulk Macro: vacuum

Thus, the structure of the device is constructed with precise coordinate and material property. In the following steps, doping, contact and mesh information will be added.

7. Add doping information

Add doping information to the bottom polygon. Choose the left and bottom polygon and double-click it. Because doping profile is defined in one polygon, it can extend to adjacent polygons; it is a good idea to input a doping block at one time.

Doping Info --> Doping Type: N

--> Set As Polygon Size

--> 1st point: (0 0 0.02)

2nd point: (10 0 0.01)

3rd point: (10 1.35 0.02)

4th point: (0 1.35 0.01)

Remember that the coordinates here are relative to the left and bottom of this polygon.

--> Max. Concentration: 1.e24

Add doping information to the top polygon. Choose the left and top polygon and double-click it:

Doping Info --> Doping Type: N

--> 1st point: 0 -0.0538 0.02)

2nd point: (50 -0.0538 0.015)

3rd point: (50 3.2962 0.02)

4th point: (0 3.2962 0.015)

--> Max. Concentration: 1.e24

8. Add contact information. Choose the left and bottom polygon and double-click it:

Contact Info --> Contact 1 --> side: double click the blank and choose 1-2
--> from: 0
--> to: 1.5

Remember that if the length of contact is longer than that of the side, the redundant length is properly ignored.

--> Type: double click the bland and choose ohmic

Repeat the steps to add two more contacts

9. Add mesh information. Choose the left and bottom polygon and double-click it:

Mesh Info --> 1st line --> Side: double click the blank and choose 1-2
--> From: 0
--> To: 1.5
--> Ratio: 0.8
--> Mesh No: 6

Repeat the steps to add all mesh information.

10. Revise the .geo , .mater and .doping files manually to get reasonable results. In this example, add mesh modification statements, for example, double_mesh, half_mesh, boundary_smooth, etc, to the .geo file.

The formed gaas2.geo is listed in the following:

```
$file:gaas2.geo
$***********GEOMETRY DESCRIPTION*******************
begin_geometry
point label=a1 xy=( 0.   0.)
point label=a2 xy=( 0.   1.4962)
point label=a3 xy=( 0.   1.5038)
point label=a4 xy=( 0.   1.7038)
point label=a5 xy=( 0.   3.0)
$
point label=b1 xy=( 1.5   0.)
point label=b2 xy=( 1.5   1.4962)
point label=b3 xy=( 1.5   1.5038)
point label=b4 xy=( 1.5   1.7038)
point label=b5 xy=( 1.5   3.0)
$
point label=c1 xy=( 8.   0.)
point label=c2 xy=( 8.   1.4962)
point label=c3 xy=( 8.   1.5038)
point label=c4 xy=( 8.   1.7038)
point label=c5 xy=( 8.   3.0)
```

```
$
$
polygon name=p1 4_points=(a1 b1 b2 a2 ) material=1 &&
   boundary_1=(a1 b1) limits_1=(0. 1.5)
polygon name=p2 4_points=(a2 b2 b3 a3 ) material=2
polygon name=p3 4_points=(a3 b3 b4 a4 ) material=3
polygon name=p4 4_points=(a4 b4 b5 a5 ) material=3   &&
   boundary_2=(a5 b5) limits_2=(0. 1.5)
polygon name=q1 4_points=(b1 c1 c2 b2 ) material=1 &&
   boundary_1=(b1 c1) limits_1=(0. 6.5)
polygon name=q2 4_points=(b2 c2 c3 b3 ) material=2
polygon name=q3 4_points=(b3 c3 c4 b4 ) material=3
polygon name=q4 4_points=(b4 c4 c5 b5 ) material=4
$
end_geometry
$***********MESH GEN******************************
begin_meshgen
put_mesh polygon=p1 edge=(b1 b2) point_from=0. point_to=1.4962 &&
   number=25 ratio=0.8 extra_point_2=0.0003
put_mesh polygon=p2 edge=(b2 b3) point_from=0. point_to=0.0076 &&
   number=6 ratio=1.0 extra_point_1=0.0003 extra_point_2=0.0003
put_mesh polygon=p3 edge=(b3 b4) point_from=0.0 point_to=0.2 &&
   number=8 ratio=1.1 extra_point_1=0.0003
put_mesh polygon=p4 edge=(b4 b5) point_from=0.0 point_to=1.2962 &&
   number=16 ratio=1.2
put_mesh polygon=p1 edge=(a1 b1) point_from=0. point_to=1.5 &&
   number=6 ratio=0.8
put_mesh polygon=q1 edge=(b1 c1) point_from=0. point_to=6.5 &&
   number=10 ratio=1.2
mesh_output   mesh_outfile=gaas2.msh order=no
$
$   If you wish to use less mesh, use order=yes above
$   and comment out the following statements.
$
$ process the mesh serveral times:
$
double_mesh   mesh_infile=gaas2.msh &&
   polygon=p1   dir=2   range1=(0.7 1.5) range2=(1.0 1.5)
double_mesh   &&
   polygon=q1   dir=2   range1=(0.0 1.0) range2=(1.0 1.5)
double_mesh   &&
   polygon=p2   dir=2   range1=(0.7 1.5)   range2=(0.0 0.008)
double_mesh   &&
   polygon=q2   dir=2   range1=0.0   1.0   range2=0.0   0.008
```

```
double_mesh   &&
   polygon=p3   dir=2   range1=0.7   1.5   range2=0.0    0.2
double_mesh   &&
   polygon=q3   dir=2   range1=0.0   1.0   range2=0.0    0.2
boundary_smooth
half_mesh   &&
   polygon=q1   dir=1   range1=(2.0 6.5) range2=(0.0 1.5)
half_mesh &&
   polygon=p1   dir=2   range1=(0.0 1.5) range2=(0.0 0.5)
$
$ Be sure to order the mesh before output
$
boundary_smooth   mesh_outfile=gaas2.msh2 order=yes
end_meshgen
```

The formed gaas2.mater is listed in the following:

```
contact type=ohmic num=1
contact type=ohmic num=2
$ ****************************
mater_var name=x1 variation=linear_y data_num=6 &&
   6_values=(0., 0.71, 1.35, 0.71, 1.4962, 0.33)
mater_var name=x3 variation=linear_y data_num=6 &&
   6_values=(1.5038, 0.33, 1.65, 0.71, 3., 0.71)
load_macro name=algaas var_name1=x1 mater=1 var_symbol1=x
load_macro name=gaas mater=2
load_macro name=algaas var_name1=x3 mater=3 var_symbol1=x
load_macro name=vacuum mater=4
get_active_layer name=AlGaAs/AlGaAs var1=0. var2=0.33 &&
   var_symbol1=xw var_symbol2=xb mater=2
active_reg type=macro thickness=0.0076 mater=2
```

The formed gaas2.doping is listed in the following:

```
doping impurity=shal_dopant charge_type=donor max_conc=1.e24   &&
   x_prof=(0.0D0, 10.0D0,   0.01D0, 0.01D0 )   &&
   y_prof= 0.D0, 1.35D0,   0.02D0, 0.02D0
doping impurity=shal_dopant charge_type=acceptor max_conc=1.e24 &&
   x_prof= 0.D0, 50.D0,   0.15D0, 0.15D0   &&
   y_prof=( 1.65,   5. , 0.02,   0.02   )
```

The formed gaas2.mplt is listed in the following:

$file:gaas2.mplt
$*************PLOTMESH*************************
begin_plotmesh
plot_mesh mesh_infile=gaas2.msh2 plot_device=postscript
end_plotmesh

The whole device is completed.

7.5.4 Example 4: Buried heterostructure laser for LASTIP

This example is provided in the path: ~\ lastip_examples \current_blocking\new_doping

This is a multiple quantum well device, functions of copy and paste are very useful in this example.

1. Create the first polygon of the device

    1. Option --> Rectangle Drawing:
        Left-click three points to define a polygon
    2. Input size and material information in popup dialogue:
        Size Info --> Height: 1.5, Width: 1
        Mater Info --> Bulk Macro: algaas
        Show Load Macro Param:x= 0.4

2. Create the second polygon

    1. Option --> Rectangle Drawing:
        Left-click the two top points of the first polygon and left-click a point upward
    2. Input size and material information in popup dialogue:
        Size Info --> Height: 0.05
        Mater Info --> Bulk Macro: algaas
        Show Load Macro Param:x= 0.2

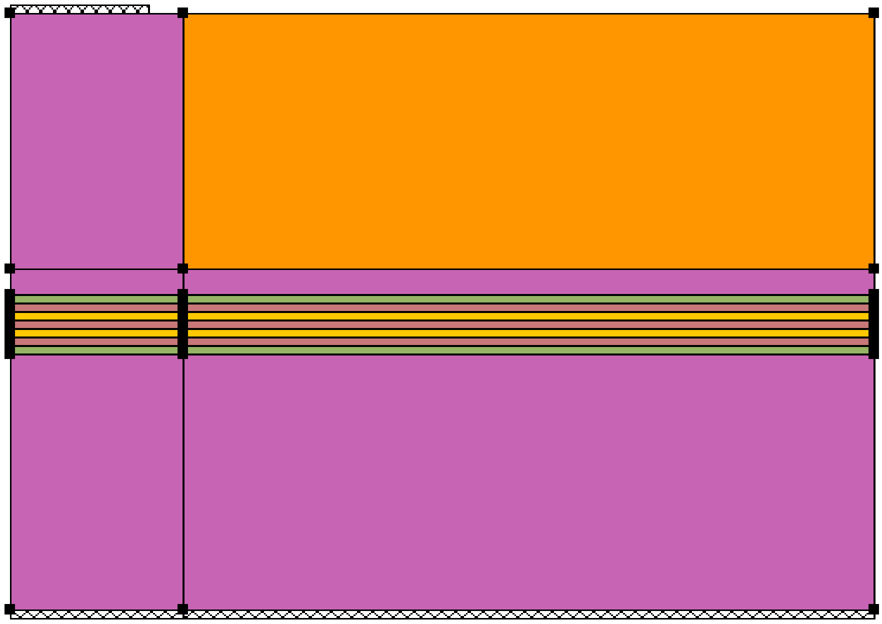

Figure 7.15: Example Newdop

3. Create the third polygon.   Repeat Step 2, except the following:

   Size Info --> Height: 0.005

   Mater Info --> Bulk Macro: gaas

4. Create the fourth polygon---active region of the device

   1. Option --> Rectangle Drawing: Left-click the two top points of the third polygon and left-click a point upward
   2. Input size and material information in popup dialogue:

   Size Info --> Height: 0.007

   Mater Info --> Bulk Macro: ingaas

   Show Load Macro Param:x= 0.185

   Active Macro: InGaAs/AlGaAs

   Show Active Macro Param --> xw: 0.185, xb: 0

5. Get fifth layer through copy and paste function

   Option --> Select: click the third polygon

   Edit --> Copy

   Edit --> Paste: click the left top point to paste a polygon on the top one directly

   To view the whole structure: Option --> Zoom Center

6. Get the left layers of the first column.   Repeat step 5 to get left layers.   In property

dialogue, modify the size and material information to get actual device structure.

7. Setup the second column by 'copy' and 'paste all'

Edit --> Select All

Edit --> Copy

Edit -> Paste: click the right bottom point to paste the whole column

Double click the second column and in popup dialogue

Size Info --> width: 4

8. Change material property of the second column. Only the top layer's material of the second

column is different from the first column in this device. Choose this polygon and double click

it. In popup dialogue:

Mater Info --> Bulk Macro: algaas

Show Load Macro Param:x= 0.6

9. Add doping information. Choose the left and bottom polygon and double-click it. Because doping

profile is defined in one polygon, it can extend to adjacent polygons, it is a good idea to input

a doping block at one time.

Doping Info --> Doping Type: N

--> Set As Polygon Size

--> 1st point: (0 0 0.02)

2nd point: (10 0 0.01)

3rd point: (10 1.5 0.02)

4th point: (0 1.5 0.01)

Remember that the coordinates here are relative to the left and bottom of this polygon.

--> Max. Concentration: 1.e24

Repeat the above steps to input the device doping information.

10. Add contact information. Choose the left and bottom polygon and double-click it:

Contact Info --> Contact 1 --> side: double click the blank and choose 1-2

--> from: 0

--> to: 1

Remember that if the length of the contact is longer than that of the side, the redundant length is properly ignored.

--> Type: double click the bland and choose ohmic

Repeat the steps to add two more contacts

11. Add mesh information. Choose the left and bottom polygon and double-click it:

Mesh Info --> 1st line --> Side: double click the blank and choose 1-2

--> From: 0

                          --> To: 1
                                    --> Ratio: 0.9
                                    --> Mesh No: 5

Repeat the steps to add all mesh information.

12. Add the information that cannot be edited by GeoEditor. Users must revise the .geo, .mater and .doping files manually.

The formed newdop.geo is listed in the following:

```
$file:newdop.geo
$
$     980nm buried newdop-wavegude laser
$       VERTICAL MESA, Al0.4GaAs clad
$          top                        :2um (1um)
$          bottom                     :2um (1um)      ________                 ________
$     yaxis-max value           :      (5um)      Al6GaAs|____              ____|
$     SCH                             :Al0.2GaAs 50nm          Al4 |_    _    _|
$     barrier                        :GaAs   5nm                        |_| |_|
$     act thickness                :7nm InGaAs                GaAs      InGaAs
$     well number                   :2
$     n-clad thickness            :1.5um
$     1st p-clad thickness     :0.15um
$     2nd p-clad thickness     :1.5um
$     buried layer                  :Al0.6GaAs
$
$***********GEOMETRY DESCRIPTION******************
begin_geometry
 point label=a1   xy= 0.    0.000
 point label=a2   xy= 0.    1.500
 point label=a3   xy= 0.    1.550
 point label=a4   xy= 0.    1.555
 point label=a5   xy= 0.    1.562
 point label=a6   xy= 0.    1.567
 point label=a7   xy= 0.    1.574
 point label=a8   xy= 0.    1.579
 point label=a9   xy= 0.    1.629
 point label=a10 xy= 0.    1.779
 point label=a11 xy= 0.    3.279
$
 point label=b1   xy= 1.    0.000
 point label=b2   xy= 1.    1.500
 point label=b3   xy= 1.    1.550
```

```
 point label=b4   xy= 1.    1.555
 point label=b5   xy= 1.    1.562
 point label=b6   xy= 1.    1.567
 point label=b7   xy= 1.    1.574
 point label=b8   xy= 1.    1.579
 point label=b9   xy= 1.    1.629
 point label=b10 xy= 1.    1.779
 point label=b11 xy= 1.    3.279
$
 point label=c1   xy= 5.    0.000
 point label=c2   xy= 5.    1.500
 point label=c3   xy= 5.    1.550
 point label=c4   xy= 5.    1.555
 point label=c5   xy= 5.    1.562
 point label=c6   xy= 5.    1.567
 point label=c7   xy= 5.    1.574
 point label=c8   xy= 5.    1.579
 point label=c9   xy= 5.    1.629
 point label=c10 xy= 5.    1.779
 point label=c11 xy= 5.    3.279
$
polygon name=p1 4_points=(a1 b1 b2 a2 ) material=1 &&
   boundary_1=(a1 b1) limits_1=(0. 1.)
polygon name=p2 4_points=(a2 b2 b3 a3 ) material=2
polygon name=p3 4_points=(a3 b3 b4 a4 ) material=3
polygon name=p4 4_points=(a4 b4 b5 a5 ) material=4
polygon name=p5 4_points=(a5 b5 b6 a6 ) material=3
polygon name=p6 4_points=(a6 b6 b7 a7 ) material=4
polygon name=p7 4_points=(a7 b7 b8 a8 ) material=3
polygon name=p8 4_points=(a8 b8 b9 a9 ) material=2
polygon name=p9 4_points=(a9 b9 b10 a10 ) material=1
polygon name=p10 4_points=(a10 b10 b11 a11 ) material=1 &&
   boundary_2=(a11 b11) limits_2=(0.2 1.)
$
polygon name=q1 4_points=(b1 c1 c2 b2 ) material=1 &&
   boundary_1=(b1 c1) limits_1=(0. 4.)
polygon name=q2 4_points=(b2 c2 c3 b3 ) material=2
polygon name=q3 4_points=(b3 c3 c4 b4 ) material=3
polygon name=q4 4_points=(b4 c4 c5 b5 ) material=4
polygon name=q5 4_points=(b5 c5 c6 b6 ) material=3
polygon name=q6 4_points=(b6 c6 c7 b7 ) material=4
polygon name=q7 4_points=(b7 c7 c8 b8 ) material=3
polygon name=q8 4_points=(b8 c8 c9 b9 ) material=2
polygon name=q9 4_points=(b9 c9 c10 b10 ) material=1
```

```
polygon name=q10 4_points=(b10 c10 c11 b11 ) material=5
$
end_geometry
$ ***********************************************
begin_meshgen
put_mesh polygon=p1 edge=(b1 b2) point_from=0. point_to=1.5 &&
   number=9 ratio=0.8
put_mesh polygon=p2 edge=(b2 b3) point_from=0.0 point_to=0.05 &&
   number=3 ratio=1.0
put_mesh polygon=p3 edge=(b3 b4) point_from=0.0 point_to=0.005 &&
   number=3 ratio=1.0
put_mesh polygon=p4 edge=(b4 b5) point_from=0.0 point_to=0.007 &&
   number=3 ratio=1.0
put_mesh polygon=p5 edge=(b5 b6) point_from=0.0 point_to=0.005 &&
   number=3 ratio=1.0
put_mesh polygon=p6 edge=(b6 b7) point_from=0.0 point_to=0.007 &&
   number=3 ratio=1.0
put_mesh polygon=p7 edge=(b7 b8) point_from=0.0 point_to=0.005 &&
   number=3 ratio=1.0
put_mesh polygon=p8 edge=(b8 b9) point_from=0.0 point_to=0.05 &&
   number=3 ratio=1.0
put_mesh polygon=p9 edge=(b9 b10) point_from=0.0 point_to=0.15 &&
   number=5 ratio=1.03
put_mesh polygon=p10 edge=(b10 b11) point_from=0.0 point_to=1.5 &&
   number=9 ratio=1.2
$
put_mesh polygon=p1 edge=(a1 b1) point_from=0. point_to=1.0 &&
   number=5 ratio=0.90
put_mesh polygon=q1 edge=(b1 c1) point_from=0. point_to=4.0 &&
   number=6 ratio=1.1
$
mesh_output mesh_outfile=newdop.msh order=yes
$
end_meshgen
```

The formed newdop.mater is listed in the following:

```
contact type=ohmic num=1
contact type=ohmic num=2
$ ****************************
load_macro name=algaas var1=0.4 mater=1 &&
```

```
   var_symbol1=x
load_macro name=algaas var1=0.2 mater=2 &&
   var_symbol1=x
load_macro name=gaas mater=3
load_macro name=ingaas var1=0.185 mater=4 &&
   var_symbol1=x
load_macro name=algaas var1=0.6 mater=5 &&
   var_symbol1=x
$
get_active_layer name=InGaAs/AlGaAs var1=0.185 var2=0. &&
   var_symbol1=xw var_symbol2=xb mater=4
active_reg type=macro mater=4 &&
     exch_coef=2.d-10 thickness=0.0070
```

The formed newdop.doping is listed in the following:

```
wave_boundary point_ll=(0., 0.1) point_ur=(5.0, 3.20) &&
   fld_center=(0.01, 1.565)

$ n-clad
doping impurity=shal_dopant charge_type=donor max_conc=1.e24   &&
   x_prof=(0.0D0, 10.0D0,   0.01D0, 0.01D0 )   &&
   y_prof=(0.0D0,   1.5D0,   0.02D0, 0.02D0 )
$ p-clad
doping impurity=shal_dopant charge_type=acceptor max_conc=1.e24   &&
   x_prof=(0.0D0    , 1.0D0,   0.15D0, 0.15D0 )   &&
   y_prof=(1.629D0, 5.0D0,   0.02D0, 0.02D0 )
doping impurity=shal_dopant charge_type=acceptor max_conc=1.e24   &&
   x_prof=(1.0D0    , 50.0D0,   0.15D0, 0.15D0 )   &&
   y_prof=(1.629D0, 1.779D0,   0.02D0, 0.02D0 )
$ n-blocking
doping impurity=shal_dopant charge_type=donor max_conc=1.e23   &&
   x_prof=(1.0D0    , 50.0D0,   0.01D0, 0.01D0 )   &&
   y_prof=(1.779D0, 5.0D0,   0.02D0, 0.02D0 )
```

The formed newdop.mplt is listed in the following:

```
$file:newdop.mplt
$*************PLOTMESH*************************
begin_plotmesh
plot_mesh mesh_infile=newdop.msh2 plot_device=postscript
end_plotmesh
```

The construction of the device is finished.

7.5.5 Example 5: Substrate radiation

This example is provided in the path: ~\lastip_examples\eeim_model\substrate_loss

This example uses a rotated version of gaas20.geo and is used to study the effect of substrate radiation. The high index GaAs is placed close to the active region. EEIM is used to solve for the radiative mode. When substrate is 0.7 um from the active region, a large radiative loss is found.

Since the radiation must be along the x-direction, we must rotate the device by 90 degrees.
Of course, we can start from gaas2 of example 3 and use the newly added functions of mirror copy and rotation provided by this version of GeoEditor to build the device model. Here, with the consideration of the new users, we still start from the very beginning.

First, we set up half of the device. Then, we use the functions of mirror copy and rotation to finish the whole device. Proficient users can also try to start from example 3 and follow the second part of this example (starting from step 11 below) to build the device. This way will save effort in building the first half of the device.

1. Create the first polygon of the device

    1. Option --> Rectangle Drawing: Left-click three points to define a polygon
    2. Input size and material information in popup dialogue:
      Size Info --> Height: 0.8, Width: 1.5
      Mater Info --> Bulk Macro: gaas
        Contact Info --> Contact 1: side 1-2, Form 0, To 1.5, Type ohmic
        Mesh Info --> Side 1-2: From 0, To 1.5, Ratio 0.8, Mesh No. 6
                  Side 2-3: From 0, To 0.8, Ratio -1.2, Mesh No. 8
    3. Click 'OK' to close this dialogue.

2. Setup the second polygon on the right side of the first one through copy and paste functions

    1. Right click the mouse button within the first polygon, click 'copy' in the popup. Then right
    click the mouse button anywhere, and click 'paste' in the popup menu. Move the newly appeared polygon to its proper position, and keep the neighbouring sides of

this polygon and the first one attached. Click the left mouse button.

2. Modify some property information of this polygon. Right click the mouse button within this polygon, click 'Property' in the popup menu.

Size Info --> Height: 0.8, Width: 6.5

Contact Info --> Contact 1: side 1-2, Form 0, To 6.5, Type ohmic

Mesh Info --> Side 1-2: From 0, To 6.5, Ratio 1.2, Mesh No. 10

No need to change other information of this polygon.

3. Click 'OK' to close this dialogue.

3. Create the third polygon on the upper side of the first one:

1. Option --> Rectangle Drawing: Left-click the two top points of the first polygon and left-click a point upward;

2. Input size and material information in popup dialogue:

Size Info --> Height: 0.6962, Width: 1.5

Mater Info --> Bulk Macro: algaas

click 'OK' in the popup dialogue

click 'x under 'Material Grading'

click 'y' under 'Direction

in the lower right table, input:

Distance 0 with Composition 0.71

Distance 0.55 with Composition 0.71

Distance 0.6962 with Composition 0.33

Mesh Info --> Side 2-3: From 0, To 0.6962, Ratio 0.8, Mesh No. 15

3. Click 'OK' to close this dialogue.

4. Setup the forth polygon on the right side of the third one through the function of material dragging

1. Option --> Rectangle Drawing: Left-click the two upper-side points of the second polygon

and left-click a point upward;

2. Click 'OK' to close this dialogue.

3. Click the 'Material Dragging (D)' on the toolbar, drag the material of the third polygon to this polygon;

5. Create the fifth polygon on the upper side of the third one:

1. Option --> Rectangle Drawing: Left-click the two top points of the third polygon and left-click a point upward

2. Input size and material information in popup dialogue:

Size Info --> Height: 0.0076, Width: 1.5

Mater Info --> Bulk Macro: gaas;

Active Macro: AlGaAs/AlGaAs

in the popup dialogue: xw 0, xb 0.33, T 300

Mesh Info --> Side 2-3: From 0, To 0.0076, Ratio 1.0, Mesh No. 6

3. Click 'OK' to close this dialogue.

6. Setup the sixth polygon on the right side of the fifth one through the function of material dragging

1. Option --> Rectangle Drawing: Left-click the two upper-side points of the forth polygon and
left-click a point upward;
2. Click 'OK' to close this dialogue.
3. Click the 'Material Dragging (D)' on the toolbar, drag the material of the fifth polygon to this polygon;

7. Create the seventh polygon on the upper side of the fifth one:

1. Option --> Rectangle Drawing: Left-click the two top points of the fifth polygon and left-click a point upward;
2. Input size and material information in popup dialogue:
Size Info --> Height: 0.2, Width: 1.5
Mater Info --> Bulk Macro: algaas
click 'OK' in the popup dialogue
click 'x under 'Material Grading'
click 'y' under 'Direction'
in the lower right table, input:
Distance 0 with Composition 0.33
Distance 0.1462 with Composition 0.71
Distance 0.2 with Composition 0.71
Mesh Info --> Side 2-3: From 0, To 0.2, Ratio 1.1, Mesh No. 8
3. Click 'OK' to close this dialogue.

8. Setup the eighth polygon on the right side of the seventh one through the function of material dragging

1. Option --> Rectangle Drawing:
Left-click the two upper-side points of the sixth polygon and
left-click a point upward
2. Click 'OK' to close this dialogue.
3. Click the 'Material Dragging(D)' on the toolbar, drag the material of the fifth polygon to this polygon

9. Create the ninth polygon on the upper side of the seventh one:

1. Option --> Rectangle Drawing: Left-click the two top points of the seventh polygon and left-click a point upward;
2. Input size and material information in popup dialogue:
   Size Info --> Height: 1.2962, Width: 1.5
   Mater Info --> Bulk Macro: algaas
   input 0.71 for x and click 'OK' in the popup dialogue;
   Contact Info --> Side 3-4: From 0, To 1.5, Type ohmic
   Mesh Info --> Side 2-3: From 0, To 1.2962, Ratio 1.2, Mesh No. 16
3. Click 'OK' to close this dialogue.

10. Create the tenth polygon on the upper side of the eighth one:

1. Option --> Rectangle Drawing: Left-click the two top points of the eighth polygon and left-click a point upward;
2. Input size and material information in popup dialogue:
   Size Info --> Height: 1.2962, Width: 6.5
   Mater Info --> Bulk Macro: vacuum
3. Click 'OK' to close this dialogue.

Now almost half of the final has been set up. The finished half is shown in Fig 7.16

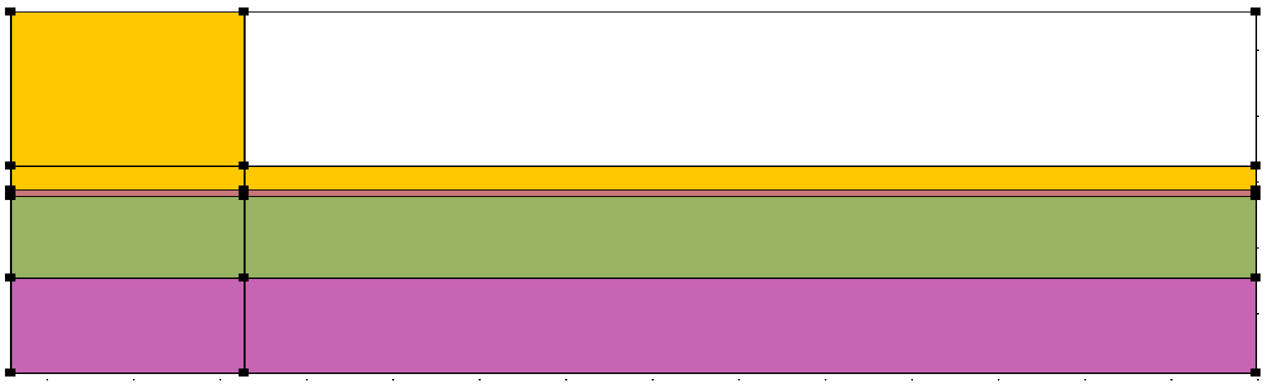

Figure 7.16: First Half of Example gaas26

Next, well use the newly available functions of 'Horizontal Mirror' and 'Rotate All Polygons' to set up the other half.

11. Set up the other half of the device by mirror copy.

1. Option --> Horizontal Mirror
Move the newly appeared mirror half to its proper position, and keep the neighbor

sides of this half and the first half attached. Click the left mouse button. This 'Horizontal Mirror' function only copies property information of size and material, so you have to add the contact and mesh information to the newly copied half of device.

2. Right-click the lower most left polygon and left click the popup menu item 'Property'. In the popup dialogue input the following information:

   Contact Info --> Contact 1: Side 1-2, From 0, To 6.5, Type ohmic

   Mesh Info --> Side 1-2, From 0, To 6.5, Ratio 0.8, Mesh No. 10

   Click 'OK' to close this dialogue.

3. Right-click the lower second left polygon and left click the popup menu item 'Property'. In the popup dialogue input the following information:

   Contact Info → Contact 1: Side 1-2, From 0, To 1.5, Type ohmic

   Mesh Info --> Side 1-2, From 0, To 1.5, Ratio 1.2, Mesh No.6

   Click 'OK' to close this dialogue.

4. Right-click the upper second left polygon and left click the popup menu item 'Property'. In the popup dialogue input the following information:

   Contact Info → Contact 1: Side 3-4, From 0, To 1.5, Type ohmic

   Click 'OK' to close this dialogue.

The device after the mirror copy and modifications in the step above are shown in Fig 7.17

12. Rotate the whole 90 degrees anti-clockwise. Click [Rotate(O)] on the toolbar. We suggest the users to examine the property information of each polygon carefully after the operation of rotation in this step to verify the rotation is what you want.

Fig 7.18 shows the device model after rotation.

13. Input the doping information.

1. Right-click the lower left polygon and left click the popup menu item 'Property'.
2. In the popup dialogue input the following information:

   Doping Info --> click 'N' under 'Doping Type'

   in the table below input:

x	y	Edge Profile
1.65	0	0.01
5	0	0.02
5	20	0.01
1.65	20	0.02

input 1.e24 for 'Max Concentration.' Click button 'Next', answer 'Y' for question "Add profile to within this polygon"

   Doping Info --> click 'P' under 'Doping Type'

   in the table below input:

x	y	Edge Profile
0	0	0.15
1.35	0	0.02

1.35	50	0.15
0	50	0.02

input 1.e24 for 'Max Concentration'

Click 'OK' to close this dialogue.

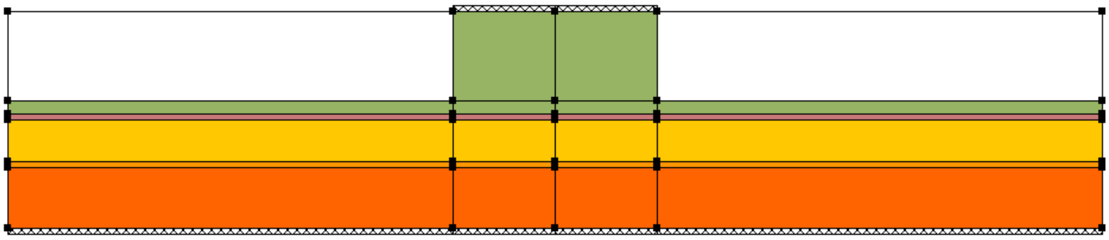

Figure 7.17: Model after Mirror Copy

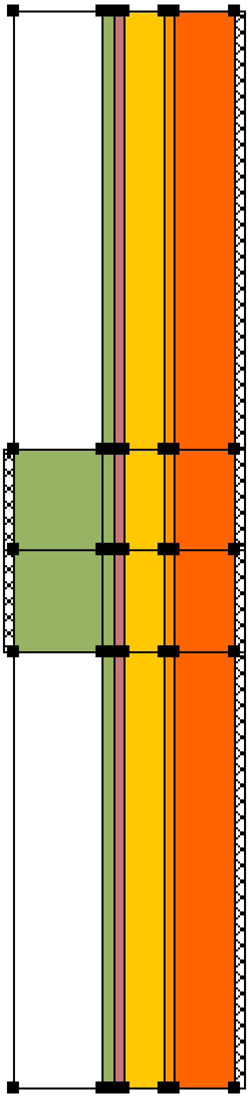

Figure 7.18: Device Model after Rotation

14. Save file as gaas26.geo, and four files of gaas26.geo, gaas26.mater, gaas26.doping and gaas26.mplt are formed.

15. Modifies files for information not available by GeoEditor. There is some information that cannot be edited by GeoEditor. Users must revise the .geo, .mater and . doping files manually to add or modify some information not available in GeoEditor. In this device, user must add put_mesh like extra_point in gaas26.geo manually as follows:

for polygon=1poly edge=[pt1 pt2], add: extra_point_2=0.0003
for polygon=2poly edge=[pt2 pt5], add: extra_point_1=0.0003
for polygon=3poly edge=[pt3 pt7], add: extra_point_2=0.0003
for polygon=5poly edge=[pt7 pt10], add: extra_point_1=0.0003
extra_point_2=0.0003
for polygon=7poly edge=[pt10 pt13], add: extra_point_1=0.0003
for polygon=11poly edge=[pt19 pt1], add: extra_point_1=0.0003
for polygon=16poly edge=[pt25 pt19], add: extra_point_2=0.0003

It is convenient to modify the gaas26.doping manually as follows:

```
doping impurity=shal_dopant charge_type=donor max_conc=1.e24   &&
   x_prof=(1.65D0, 5.0D0,   0.02D0, 0.02D0)     &&
   y_prof=(0.0D0, 20.0D0,   0.01D0, 0.01D0 )
doping impurity=shal_dopant charge_type=acceptor max_conc=1.e24 &&
   x_prof=( 0.0,   1.35 , 0.02,   0.02   )   &&
   y_prof=( 0.D0, 50.D0,   0.15D0, 0.15D0)
```

The formed gaas26.geo is listed in the following:

```
$file:************gaas26.geo************

$***********GEOMETRY DESCRIPTION*******************
begin_geometry
point label=pt4 xy=[ 2.200000 8.000000 ]
point label=pt1 xy=[ 3.000000 8.000000 ]
point label=pt2 xy=[ 3.000000 9.500000 ]
point label=pt3 xy=[ 2.200000 9.500000 ]
point label=pt5 xy=[ 3.000000 16.000000 ]
point label=pt6 xy=[ 2.200000 16.000000 ]
point label=pt8 xy=[ 1.503800 8.000000 ]
point label=pt7 xy=[ 1.503800 9.500000 ]
point label=pt9 xy=[ 1.503800 16.000000 ]
point label=pt11 xy=[ 1.496200 8.000000 ]
```

```
point label=pt10 xy=[ 1.496200 9.500000 ]
point label=pt12 xy=[ 1.496200 16.000000 ]
point label=pt14 xy=[ 1.296200 8.000000 ]
point label=pt13 xy=[ 1.296200 9.500000 ]
point label=pt15 xy=[ 1.296200 16.000000 ]
point label=pt17 xy=[ 0.000000 8.000000 ]
point label=pt16 xy=[ 0.000000 9.500000 ]
point label=pt18 xy=[ 0.000000 16.000000 ]
point label=pt20 xy=[ 2.200000 6.500000 ]
point label=pt19 xy=[ 3.000000 6.500000 ]
point label=pt21 xy=[ 1.503800 6.500000 ]
point label=pt22 xy=[ 1.496200 6.500000 ]
point label=pt23 xy=[ 1.296200 6.500000 ]
point label=pt24 xy=[ 0.000000 6.500000 ]
point label=pt26 xy=[ 2.200000 0.000000 ]
point label=pt25 xy=[ 3.000000 0.000000 ]
point label=pt27 xy=[ 1.503800 0.000000 ]
point label=pt28 xy=[ 1.496200 0.000000 ]
point label=pt29 xy=[ 1.296200 0.000000 ]
point label=pt30 xy=[ 0.000000 0.000000 ]
polygon name=1poly 4_points=[pt4 pt1 pt2 pt3 ] material=1 &&
   boundary_1=[ pt1 pt2 ] limits_1=[ 0.00000E+000   1.50000E+000 ]
polygon name=2poly 4_points=[pt3 pt2 pt5 pt6 ] material=1 &&
   boundary_1=[ pt2 pt5 ] limits_1=[ 0.00000E+000   6.50000E+000 ]
polygon name=3poly 4_points=[pt8 pt4 pt3 pt7 ] material=2
polygon name=4poly 4_points=[pt7 pt3 pt6 pt9 ] material=2
polygon name=5poly 4_points=[pt11 pt8 pt7 pt10 ] material=3
polygon name=6poly 4_points=[pt10 pt7 pt9 pt12 ] material=3
polygon name=7poly 4_points=[pt14 pt11 pt10 pt13 ] material=4
polygon name=8poly 4_points=[pt13 pt10 pt12 pt15 ] material=4
polygon name=9poly 4_points=[pt17 pt14 pt13 pt16 ] material=4 &&
   boundary_2=[ pt16 pt17 ] limits_2=[ 0.00000E+000   1.50000E+000 ]
polygon name=10poly 4_points=[pt16 pt13 pt15 pt18 ] material=5
polygon name=11poly 4_points=[pt20 pt19 pt1 pt4 ] material=1 &&
   boundary_1=[ pt19 pt1 ] limits_1=[ 0.00000E+000   1.50000E+000 ]
polygon name=12poly 4_points=[pt21 pt20 pt4 pt8 ] material=2
polygon name=13poly 4_points=[pt22 pt21 pt8 pt11 ] material=3
polygon name=14poly 4_points=[pt23 pt22 pt11 pt14 ] material=4
polygon name=15poly 4_points=[pt24 pt23 pt14 pt17 ] material=4 &&
   boundary_2=[ pt17 pt24 ] limits_2=[ 0.00000E+000   1.50000E+000 ]
polygon name=16poly 4_points=[pt26 pt25 pt19 pt20 ] material=1 &&
   boundary_1=[ pt25 pt19 ] limits_1=[ 0.00000E+000   6.50000E+000 ]
polygon name=17poly 4_points=[pt27 pt26 pt20 pt21 ] material=2
polygon name=18poly 4_points=[pt28 pt27 pt21 pt22 ] material=3
```

```
polygon name=19poly 4_points=[pt29 pt28 pt22 pt23 ] material=4
polygon name=20poly 4_points=[pt30 pt29 pt23 pt24 ] material=5
$

end_geometry
$***********MESH GEN*****************************
begin_meshgen
put_mesh   polygon=1poly   edge=[ pt1 pt2] &&
       point_from= 0.000000e+000   point_to=1.500000e+000 &&
       number=6 ratio=8.000000e-001   extra_point_2=0.0003
put_mesh   polygon=1poly   edge=[ pt2 pt3] &&
       point_from= 0.000000e+000   point_to=8.000000e-001 &&
       number=8 ratio=-1.200000e+000
put_mesh   polygon=2poly   edge=[ pt2 pt5] &&
       point_from= 0.000000e+000   point_to=6.500000e+000 &&
       number=10 ratio=1.200000e+000   extra_point_1=0.0003
put_mesh   polygon=3poly   edge=[ pt3 pt7] &&
       point_from= 0.000000e+000   point_to=6.962000e-001 &&
       number=15 ratio=8.000000e-001   extra_point_2=0.0003
put_mesh   polygon=5poly   edge=[ pt7 pt10] &&
       point_from= 0.000000e+000   point_to=7.600000e-003 &&
       number=6 ratio=1.000000e+000 &&
        extra_point_1=0.0003 extra_point_2=0.0003
put_mesh   polygon=7poly   edge=[ pt10 pt13] &&
       point_from= 0.000000e+000   point_to=2.000000e-001 &&
       number=8 ratio=1.100000e+000   extra_point_1=0.0003
put_mesh   polygon=9poly   edge=[ pt13 pt16] &&
       point_from= 0.000000e+000   point_to=1.296200e+000 &&
       number=16 ratio=1.200000e+000
put_mesh   polygon=11poly   edge=[ pt19 pt1] &&
       point_from= 0.000000e+000   point_to=1.500000e+000 &&
       number=6 ratio=1.200000e+000   extra_point_1=0.0003
put_mesh   polygon=16poly   edge=[ pt25 pt19] &&
       point_from= 0.000000e+000   point_to=6.500000e+000 &&
       number=10 ratio=8.000000e-001   extra_point_2=0.0003
mesh_output mesh_outfile=gaas26.msh      order=yes
end_meshgen
```

The formed gaas26.mater is listed in the following:

```
$file:*************gaas26.mater*************

contact num= 1 type=ohmic
```

```
contact num= 2 type=ohmic
load_macro name= gaas mater= 1
mater_var name= c2 variation= linear_x data_num= 6 &&
6_values=[   &&
      1.5038E+000 3.3000E-001 &&
      1.6500E+000 7.1000E-001 &&
      2.2000E+000 7.1000E-001]
load_macro name= algaas mater= 2 var_name1= c2
load_macro name= gaas mater= 3
get_active_layer name= AlGaAs/AlGaAs mater= 3   &&
      var1= 0.000E+000 &&
      var2= 3.300E-001 &&
      var3= 3.000E+002
active_reg type=macro mater= 3 &&
      thickness=7.60000E-003
mater_var name= c4 variation= linear_x data_num= 8 &&
8_values=[   &&
      0.0000E+000 7.1000E-001 &&
      1.2962E+000 7.1000E-001 &&
      1.3500E+000 7.1000E-001 &&
      1.4962E+000 3.3000E-001]
load_macro name= algaas mater= 4 var_name1= c4
load_macro name= vacuum mater= 5
```

The formed gaas26.doping is listed in the following:

```
doping impurity=shal_dopant charge_type=donor max_conc=1.e24   &&
   x_prof=(1.65D0, 5.0D0,   0.02D0, 0.02D0)     &&
   y_prof=(0.0D0, 20.0D0,   0.01D0, 0.01D0 )
doping impurity=shal_dopant charge_type=acceptor max_conc=1.e24 &&
   x_prof=( 0.0,   1.35 , 0.02,   0.02   )   &&
   y_prof=( 0.D0, 50.D0,   0.15D0, 0.15D0)
```

The formed gaas26.mplt is listed in the following:

```
$file:gaas26.mplt
$*************PLOTMESH**************************
begin_plotmesh
plot_mesh mesh_infile=gaas26.msh plot_device=window
end_plotmesh
```

The whole device is completed.